

DOKUMEN PENDOKUMENTASIAN HAK CIPTA PROGRAM KOMPUTER (DJKI KEMENKUMHAM RI)

USER MANUAL, ARSITEKTUR, APIS, & KODE SUMBER PER FITUR

Aplikasi PeatFR — Fire Intelligence System & Early Warning Kebakaran Lahan Gambut

(Mahdiyasa et al., 2025)

Data Pendaftaran Hak Cipta (HKI)

Parameter Hak Cipta (HKI)	Keterangan Data Resmi Pendaftaran
Jenis Ciptaan	**Program Komputer** (Aplikasi Web Telemetri GIS Satelit & AI Fire Forecasting)
Judul Ciptaan	**Aplikasi PeatFR — Fire Intelligence System & Early Warning Kebakaran Lahan Gambut**
Pencipta & Pemegang Hak Cipta	**Bagas Wibowo, S.Pd., M.Eng.** & Tim Peneliti Kelompok Keahlian SEAL Telkom University
Tanggal & Tempat Pertama Diumumkan	**31 Agustus 2026** di Bandung, Jawa Barat, Republik Indonesia
Teknologi Stack	**Python 3.13** (FastAPI, PyTorch, SciPy) & **React 18** (TypeScript, Vite, Leaflet GIS)
URL Akses Sistem	https://peatfr.bagaswibowo.app/

1. GAMBARAN UMUM APLIKASI

PeatFR (Peatland Fire Vulnerability Index & Early Warning System) adalah platform pemantauan dan peringatan dini berbasis web yang dirancang khusus untuk memetakan dan memprediksi risiko kebakaran lahan gambut tropis di Indonesia (Mahdiyasa et al., 2025 — *Ecological Informatics* 92, DOI 10.1016/j.ecoinf.2025.103532).

Aplikasi ini menghubungkan empat lapis data:

- Telemetri lingkungan realtime** — Open-Meteo / ERA5-Land (suhu maksimum, soil moisture 0–7 cm & 7–28 cm, presipitasi harian).
- Citra thermal satelit** — NASA FIRMS (VIIRS 375m SNPP/NOAA-20/NOAA-21 & MODIS) melalui MAP_KEY terotorisasi, plus overlay WMS NASA GIBS.
- Vector alert kebakaran gambut** — Global Forest Watch (GFW) untuk deteksi titik api spesifik di poligon lahan gambut Indonesia, serta agregasi FirePing (deteksi terdekat & area terbakar GWIS).
- Model kecerdasan buatan** — ARIMA (stokastik + Box-Cox), LSTM & GRU (PyTorch) untuk memproyeksikan Indeks Kerawanan Kebakaran Gambut (PFVI) hingga 14 hari ke depan, dikalibrasi dengan optimasi Nelder-Mead (SciPy).

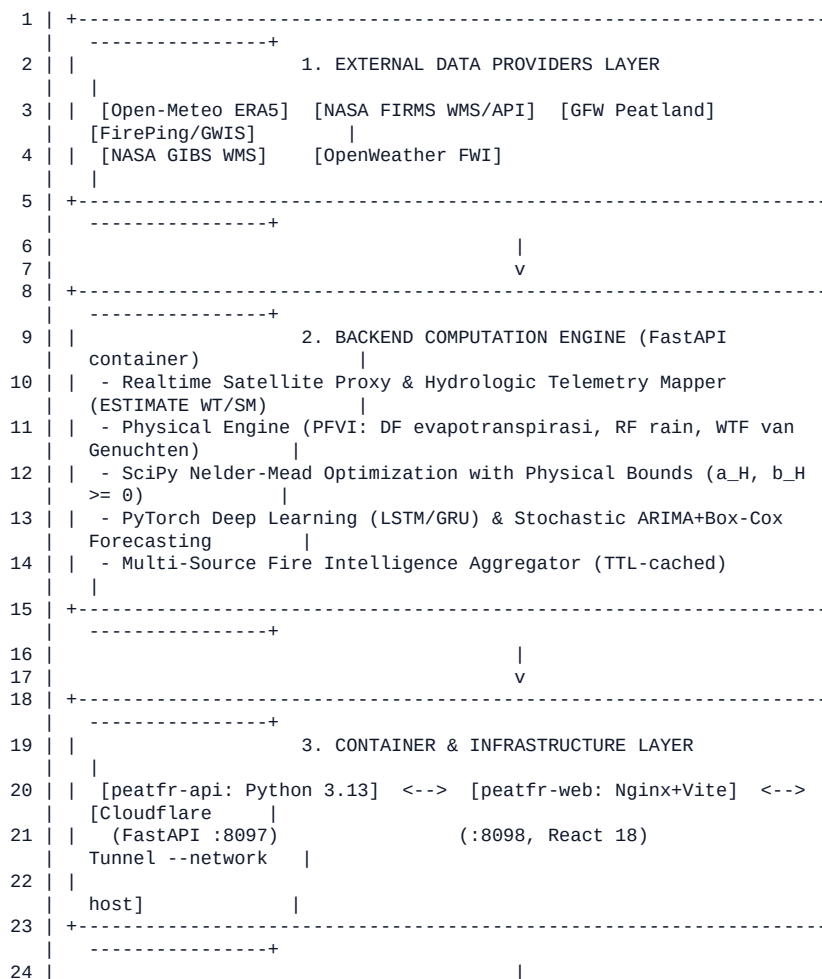
Seluruh aplikasi berjalan dalam arsitektur container Docker (peatfr-api, peatfr-web, cftunnel-peatfr) dan dipublikasikan melalui Cloudflare Tunnel ke domain <https://peatfr.bagaswibowo.app/>.

2. TUJUAN UTAMA APLIKASI

1. **Deteksi dini kebakaran bawah permukaan (smoldering fire):** memantau penurunan Muka Air Tanah (TMA) hingga di bawah batas kritis regulasi PP No. 71/2014 (-0.4 m), karena kebakaran gambut tropis dominan terjadi pada lapisan akrotelm yang kering.
1. **Kuantifikasi kerawanan dengan indeks fisik terkalibrasi (PFVI):** mensimulasikan neraca air harian (evapotranspirasi DF, curah hujan efektif RF, kapilaritas tanah WTF) dan mengkalibrasi parameter hidrologi (a_H , b_H , n , α) terhadap Drought Index observasi dari soil moisture.
1. **Proyeksi multi-model AI:** memberikan forecast 4/7/14 hari ke depan menggunakan ARIMA + Box-Cox, LSTM, dan GRU, dengan kategori risiko Low (<75), Moderate ($75-150$), High ($150-225$), Extreme (≥ 225).
1. **Analisis lanjutan untuk peneliti/pakar (Mode Pakar):** kontrol penuh atas metode imputasi data missing (kNN, spline, LOESS, linear) dan hyperparameter model.
1. **Operasional mitigasi berbasis bukti:** peta satelit live interaktif, simulasi skenario "what-if", dan data fire-intelligence multi-sumber untuk koordinasi lapangan BPBD/Manggala Agni.

3. ARSITEKTUR SISTEM & DESAIN TEKNIS (SYSTEM ARCHITECTURE)

Sistem PeatFR dirancang menggunakan arsitektur 4-tier:



```
25 |                                     v
26 | +-----+
27 | |                                     4. CLIENT PRESENTATION LAYER (React 18 +
28 | | Leaflet) |
29 | | - Header Navigasi & Pilih Lokasi Provinsi/Kabupaten
30 | | - Risk Status Gauge & Fire Intelligence Cockpit (PFVI, FirePing,
31 | | FWI, GWIS) |
32 | | - Live GIS Satellite Map (NASA FIRMS WMS, VIIRS hotspots, GFW,
33 | | GIBS, klik lokasi)|
34 | | - Pipeline Controls (imputasi, model AI, horizon) + Forecast
35 | | Charts |
36 | | - What-If Scenario Simulator & Spesifikasi Teoretis (Modal)
37 | |
38 | +-----+
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99 | +-----+
100 | +-----+
```

Persamaan Matematika Model Fisika PeatFR:

***Laju Evapotranspirasi (DF_t):**

$$DF_t = [(300 - PFVI_{t-1}) \cdot (0.4982 \cdot e^{0.0905 \cdot T_{max} + 1.6096} - 4.268) \cdot dt \cdot 10^{-3}] / [1 + 10.88 \cdot e^{-0.0017358 \cdot R_0}]$$

- **Faktor Curah Hujan Efektif (RF_t):** $RF_t = \max(0, R_f_t - 5.1)$ (interception kanopi 5.1 mm/hari)
- **Faktor Muka Air Tanah (WTF_t, van Genuchten):** $WTF_t = a_H - b_H \cdot (1 - \theta(v)) \cdot 300$, dengan $\theta(v) = [1 + (v/\alpha)^n]^{1-1/n}$, v = kedalaman TMA (cm)
- **Drought Index Observasi (DI_obs):** $DI = 300 \cdot (1 - (SM - fc)/(sat - fc))$, di-clip ke $[0, 300]$
- **Kendala Fisik Optimasi:** $a_H \geq 0, b_H \geq 0, n > 1.05, \alpha > 0.01$

4. DOKUMENTASI API & SUMBER DATA (DATA SOURCES & APIS)

A. Sumber Data Satelit Eksternal

Provider & Dataset	Endpoint / Service URL	Variabel & Penggunaan
Open-Meteo / ERA5-Land	<code>`https://api.open-meteo.com/v1/forecast`</code>	Suhu max 2m, soil moisture (0–7cm, 7–28cm), presipitasi harian; timezone Asia/Jakarta
NASA FIRMS Area API	<code>`https://firms.modaps.eosdis.nasa.gov/api/area/csv/{MAP_KEY}/{source}/{bbox}/{days}`</code>	Hotspot VIIRS 375m & MODIS: lat/lon, bright_ti4 (K), FRP (MW), confidence, daynight
NASA FIRMS WMS	<code>`https://firms.modaps.eosdis.nasa.gov/maps/server/wms/`</code>	Layer thermal VIIRS 24j & MODIS 24j (MAP_KEY terotorisasi)
NASA GIBS WMS	<code>`https://gibs.earthdata.nasa.gov/wms/epsg3857/`</code>	Citra thermal anomalies VIIRS SNPP 375m pass siang & malam
GFW Peatland Alerts	<code>`https://data-api.globalforestwatch.org/dataset/nasa_viirs_fire_alerts/latest/query`</code>	Query SQL: <code>`iso='IDN' AND is_peatland=true`</code> — kebakaran pada lahan gambut
FirePing Public API	<code>`https://fireping.net/api/public/v1/fires/near`</code>	Deteksi api terdekat (≤ 25 km), jarak, burned area GWIS 7-hari (zero-key)
OpenWeather FWI	<code>`https://api.openweathermap.org/data/2.5/fwi/forecast`</code>	Fire Weather Index (opsional; fallback ke estimasi meteorologis internal)

B. Spesifikasi Internal REST API (FastAPI Engine — container `peatfr-api:8097`)

HTTP Method & Route	Fungsi & Deskripsi Operasional
`GET /`	Root service info (status, service name, version, docs link)
`GET /api/v1/health`	Health check engine (status: ok)
`GET /api/v1/indonesia/regions`	Hirarki administrasi provinsi & kabupaten (JSON `indonesia_admin.json`)
`GET /api/v1/realtime-peatland-data?lat&lon;&days;`	Proxy telemetri Open-Meteo → deret WT/SM/Rf/Temp + fire intelligence (FirePing + FWI)
`GET /api/v1/fire-intelligence/nearby?lat&lon;&radius;_m`	Deteksi titik api terdekat & burned area (FirePing/GWIS)
`GET /api/v1/fire-intelligence/gfw-peatland-fires?limit`	Vector kebakaran gambut GFW (is__peatland=true, iso=IDN)
`GET /api/v1/fire-intelligence/firms-hotspots?bbox&source;&day;_range`	Hotspot NASA FIRMS via MAP_KEY (VIIRS_SNPP_NRT, MODIS_NRT, VIIRS_NOAA20_NRT, VIIRS_NOAA21_NRT)
`GET /api/v1/fire-intelligence/severe-alerts`	Peringatan area kebakaran gambut parah Indonesia
`GET /api/v1/fire-intelligence/fwi?lat&lon;`	Fire Weather Index (OpenWeather / estimasi meteorologis)
`GET /api/v1/sabangau-sample`	Dataset sampel 192 hari Sabangau (Kalteng) dengan missing data intrinsik
`POST /api/v1/impute`	Imputasi data missing: kNN, spline, LOESS, linear
`POST /api/v1/forecast`	Forecasting ARIMA/LSTM/GRU (h, look_back, hidden_units, epochs)
`POST /api/v1/pipeline/auto`	Pipeline penuh: imputasi → forecasting → optimasi Nelder-Mead → simulasi PFVI → kategori risiko

5. PANDUAN PETUNJUK FITUR-DEMI-FITUR & KODE SUMBER TERHUBUNG

FITUR 1: Header Navigasi & Pemilih Lokasi (Provinsi/Kabupaten + Tombol Aksi)

Deskripsi & Tujuan Fitur: Komponen navigasi utama aplikasi. Menampilkan brand PeatFR dengan badge 'Fire Intelligence Light', informasi jurnal (Mahdiyasa et al., 2025), dropdown Provinsi & Kabupaten/Kota (dengan label '(Gambut)' untuk wilayah lahan gambut), tombol **Satelit Realtime** untuk memuat ulang data satelit, dan tombol **Spesifikasi Teoretis** untuk membuka modal metodologi jurnal.

Cara Menggunakan:

- Pilih **Provinsi** pada dropdown pertama (default: Riau).
- Pilih **Kabupaten/Kota** pada dropdown kedua (default: Kab. Siak — ditandai '(Gambut)').
- Tekan tombol **Satelit Realtime** untuk menarik ulang data Open-Meteo & fire intelligence lokasi terpilih.
- Tekan tombol **Spesifikasi Teoretis** untuk melihat metadata jurnal PeatFR dan daftar integrasi API satelit.

Kode Sumber (Source Code Listing): frontend/src/components/Header.tsx

```

1 | import React, { useState, useEffect } from 'react';
2 | import axios from 'axios';
3 | import { Flame, RefreshCw, BookOpen } from 'lucide-react';
4 |
5 | export interface Regency {
6 |   id: string;
7 |   name: string;
8 |   lat: number;
9 |   lon: number;
10 |   peat: boolean;
11 | }
12 |
13 | export interface Province {
14 |   id: string;
15 |   name: string;
16 |   lat: number;
17 |   lon: number;
18 |   regencies: Regency[];
19 | }
20 |
21 | interface HeaderProps {
22 |   selectedProvince: Province | null;
23 |   selectedRegency: Regency | null;
24 |   onSelectRegion: (prov: Province, reg: Regency) => void;
25 |   onLoadRealtimeData: () => void;
26 |   onOpenPaperModal: () => void;
27 |   loadingRealtime: boolean;
28 | }
29 |
30 | export const Header: React.FC<HeaderProps> = ({
31 |   selectedProvince,
32 |   selectedRegency,
33 |   onSelectRegion,
34 |   onLoadRealtimeData,
35 |   onOpenPaperModal,
36 |   loadingRealtime
37 | }) => {
38 |   const [provinces, setProvinces] = useState<Province[]>([]);
39 |
40 |   useEffect(() => {
41 |     axios.get('/api/v1/indonesia/regions')
42 |       .then((res) => {
43 |         if (res.data && res.data.provinces) {
44 |           setProvinces(res.data.provinces);
45 |         }
46 |       })
47 |       .catch((err) => console.warn('Could not fetch Indonesia
48 | regions:', err));
49 |   }, []);
50 |
51 |   return (
52 |     <header className="border-b border-slate-200 bg-white px-4
53 | lg:px-8 py-3.5 mb-6 shadow-sm">
54 |       <div className="max-w-7xl mx-auto flex flex-col md:flex-row
55 | items-center justify-between gap-4">
56 |         { /* Brand & Subtitle */ }
57 |         <div className="flex items-center gap-3 w-full md:w-auto
58 | justify-between md:justify-start">
59 |           <div className="flex items-center gap-3">
60 |             <div className="p-2.5 rounded-lg bg-red-50 border
61 | border-red-200 text-red-600">
62 |               <Flame className="w-5 h-5" />
63 |             </div>
64 |             <div>
65 |               <div className="flex items-center gap-2">
66 |                 <span className="text-lg font-bold text-slate-900
67 | tracking-tight">PeatFR</span>
68 |                 <span className="text-slate-400"></span>
69 |                 <span className="text-xs font-mono font-bold
70 | text-emerald-700 bg-emerald-50 px-2 py-0.5 rounded border
71 | border-emerald-200">
72 |                   Fire Intelligence Light
73 |                 </span>
74 |               </div>
75 |               <p className="text-[11px] text-slate-500 font-medium">
76 |                 Sistem Peringatan Dini Kebakaran Lahan Gambut
77 |                 Indonesia (Mahdiyasa et al., 2025)
78 |               </p>
79 |             </div>
80 |           </div>
81 |         </div>
82 |       </div>
83 |     </header>
84 |   );
85 | }

```

```

73 |
74 |     { /* Location Dropdowns & Actions */ }
75 |     <div className="flex flex-wrap items-center gap-2.5 w-full
76 |         md:w-auto justify-end">
77 |         { /* Province Selector */ }
78 |         <select
79 |             value={selectedProvince?.id || ''}
80 |             onChange={(e) => {
81 |                 const foundProv = provinces.find((p) => p.id ===
82 |                 e.target.value);
83 |                 if (foundProv && foundProv.regencies.length > 0) {
84 |                     onSelectRegion(foundProv, foundProv.regencies[0]);
85 |                 }
86 |             }}
87 |             className="bg-slate-50 border border-slate-300
88 |             rounded-md px-3 py-1.5 text-xs font-mono text-slate-800
89 |             focus:outline-none focus:border-emerald-600 cursor-pointer
90 |             font-medium"
91 |             >
92 |             {provinces.map((prov) => (
93 |                 <option key={prov.id} value={prov.id}
94 |                 className="bg-white text-slate-900">
95 |                     {prov.name}
96 |                 </option>
97 |             ))}
98 |         </select>
99 |
100 |         { /* Regency Selector */ }
101 |         <select
102 |             value={selectedRegency?.id || ''}
103 |             onChange={(e) => {
104 |                 if (selectedProvince) {
105 |                     const foundReg = selectedProvince.regencies.find((r)
106 |                     => r.id === e.target.value);
107 |                     if (foundReg) onSelectRegion(selectedProvince,
108 |                     foundReg);
109 |                 }
110 |             }}
111 |             className="bg-slate-50 border border-slate-300
112 |             rounded-md px-3 py-1.5 text-xs font-mono text-slate-800
113 |             focus:outline-none focus:border-emerald-600 cursor-pointer
114 |             font-medium"
115 |             >
116 |             {selectedProvince?.regencies.map((reg) => (
117 |                 <option key={reg.id} value={reg.id}
118 |                 className="bg-white text-slate-900">
119 |                     {reg.name} {reg.peat ? '(Gambut)' : ''}
120 |                 </option>
121 |             ))}
122 |         </select>
123 |
124 |         <button
125 |             onClick={onLoadRealtimeData}
126 |             disabled={loadingRealtime}
127 |             className="flex items-center gap-1.5 px-3 py-1.5 text-xs
128 |             font-semibold rounded-md bg-emerald-600 hover:bg-emerald-700
129 |             text-white transition-colors disabled:opacity-50 shadow-sm"
130 |             >
131 |             <RefreshCw className={`w-3.5 h-3.5 ${loadingRealtime ?
132 |             'animate-spin' : ''}`} />
133 |             <span>Satelit Realtime</span>
134 |         </button>
135 |
136 |         <button
137 |             onClick={onOpenPaperModal}
138 |             className="flex items-center gap-1.5 px-3 py-1.5 text-xs
139 |             font-semibold rounded-md bg-slate-100 hover:bg-slate-200
140 |             text-slate-700 border border-slate-300 transition-colors"
141 |             >
142 |             <BookOpen className="w-3.5 h-3.5 text-slate-600" />
143 |             <span>Spesifikasi Teoretis</span>
144 |         </button>
145 |     </div>
146 | </div>
147 | </header>
148 | );
149 | };

```

Screenshot Aplikasi:

[Screenshot: Header PeatFR: navigasi, pilih lokasi, dan tombol aksi — ss_header.png]

FITUR 2: Cockpit Status Kerawanan (Risk Status Gauge) & Fire Intelligence

Deskripsi & Tujuan Fitur: Panel telemetri atas menampilkan nilai **Indeks Kerawanan Kebakaran Gambut (PFVI)** saat ini (0–300) dengan badge kategori risiko (LOW/MODERATE/HIGH/EXTREME), progress bar severity, hasil optimasi **Nelder-Mead** (a_H , b_H , n , α , MSE), tiga kartu fire intelligence (**FirePing Proximity** — jarak titik api terdekat & jumlah deteksi 24 jam; **GWIS 7-Day Burned Area** — estimasi area terbakar; **FWI** — Fire Weather Index & danger rating), serta empat panel telemetri lingkungan (**WT** muka air tanah, **SM** soil moisture, **Rf** curah hujan, **Temp** suhu maksimum) dengan ambang kritis masing-masing.

Cara Menggunakan:

1. Amati angka PFVI besar dan badge warna untuk status kerawanan saat ini (proyeksi h-hari ke depan).
1. Cek blok **Nelder-Mead Opt** untuk melihat parameter kalibrasi hidrologi hasil optimasi (MSE semakin kecil semakin baik).
1. Periksa tiga kartu fire intelligence: jarak titik api terdekat (km), area terbakar (ha), dan skor FWI beserta danger rating.
1. Pantau empat kolom telemetri: WT < -0.8 m = Kritis; SM < 45% = Tanah Kering; Rf < 5.1 mm = Tanpa Hujan Efektif; Temp > 34.5 °C = Suhu Tinggi.

Kode Sumber (Source Code Listing): frontend/src/components/RiskStatusGauge.tsx

```

1 | import React from 'react';
2 |
3 | export interface FireIntelligenceData {
4 |   nearby?: {
5 |     status?: string;
6 |     detection_count?: number;
7 |     nearest_distance_km?: number | null;
8 |     latest_detection_at?: string;
9 |     burned_area_ha?: number;
10 |    satellite_sources?: string[];
11 |   };
12 |   fwi?: {
13 |     status?: string;
14 |     source?: string;
15 |     fwi_score?: number;
16 |     danger_rating?: string;
17 |   };
18 | }
19 |
20 | interface RiskStatusGaugeProps {
21 |   pfvi: number;
22 |   status: string;
23 |   waterTable: number;
24 |   soilMoisture: number;
25 |   rainfall: number;
26 |   temp: number;
27 |   forecastDays: number;
28 |   minPfvi?: number;
29 |   maxPfvi?: number;
30 |   fireIntelligence?: FireIntelligenceData;
31 |   optimizedParams?: {
32 |     a_h: number;
33 |     b_h: number;
34 |     n: number;
35 |     alpha: number;
36 |     mse: number;
37 |   };
38 | }
39 |
40 | export const RiskStatusGauge: React.FC<RiskStatusGaugeProps> = ({
41 |   pfvi,
42 |   status,
43 |   waterTable,
44 |   soilMoisture,
45 |   rainfall,
46 |   temp,
47 |   forecastDays,
48 |   minPfvi = 45.0,
49 |   maxPfvi = 285.0,
50 |   fireIntelligence,

```

```

51 |   optimizedParams
52 | }) => {
53 |   const getBadgeStyle = (category: string) => {
54 |     switch (category.toLowerCase()) {
55 |       case 'low':
56 |         return 'bg-emerald-100 text-emerald-800 border-emerald-300';
57 |       case 'moderate':
58 |         return 'bg-amber-100 text-amber-800 border-amber-300';
59 |       case 'high':
60 |         return 'bg-orange-100 text-orange-800 border-orange-300';
61 |       case 'extreme':
62 |       case 'very high':
63 |         return 'bg-red-100 text-red-800 border-red-300';
64 |       default:
65 |         return 'bg-slate-100 text-slate-800 border-slate-300';
66 |     }
67 |   };
68 |
69 |   const pfviPercentage = Math.min(100, Math.max(0, (pfvi / 300) *
100 |     100));
101 |
102 |   const nearby = fireIntelligence?.nearby;
103 |   const fwi = fireIntelligence?.fwi;
104 |
105 |   return (
106 |     <div className="space-y-4 mb-6">
107 |       {/* Top Telemetry Header Panel */}
108 |       <div className="telemetry-panel rounded-xl p-5 bg-white border
109 |         border-slate-200 shadow-sm">
110 |         <div className="flex flex-col lg:flex-row items-start
111 |           lg:items-center justify-between gap-6 pb-5 border-b
112 |           border-slate-200">
113 |           <div>
114 |             <div className="text-xs font-mono font-bold uppercase
115 |               tracking-wider text-slate-500 mb-1">
116 |               Indeks Kerawanan Kebakaran Gambut (PFVI)
117 |             </div>
118 |             <div className="flex items-baseline gap-3">
119 |               <span className="text-4xl font-mono font-extrabold
120 |                 text-slate-900 tracking-tight">
121 |                 {pfvi.toFixed(1)}
122 |               </span>
123 |               <span className="text-xs font-mono text-slate-500
124 |                 font-semibold">/ 300.0</span>
125 |               <span className={`px-2.5 py-0.5 text-xs font-mono
126 |                 font-bold rounded border ${getBadgeStyle(status)}`>
127 |                 {status.toUpperCase()}
128 |               </span>
129 |             </div>
130 |             <p className="text-xs text-slate-500 font-medium mt-1">
131 |               Proyeksi {forecastDays} Hari Ke Depan (Rentang 30
132 |               Hari: {minPfvi.toFixed(1)} s.d {maxPfvi.toFixed(1)})
133 |             </p>
134 |           </div>
135 |           {/* Nelder-Mead Parameters Readout */}
136 |           {optimizedParams && (
137 |             <div className="bg-slate-50 p-3 rounded-lg border
138 |               border-slate-200 text-xs font-mono space-y-1">
139 |               <div className="text-[11px] text-slate-700 font-bold
140 |                 uppercase tracking-wider flex items-center justify-between gap-4">
141 |                 <span>Nelder-Mead Opt</span>
142 |                 <span className="text-slate-500">MSE:
143 |                 {optimizedParams.mse.toFixed(2)}</span>
144 |               </div>
145 |               <div className="text-slate-600 grid grid-cols-2
146 |                 gap-x-4 gap-y-0.5 text-[11px]">
147 |                 <span>a_H = {optimizedParams.a_h.toFixed(2)}</span>
148 |                 <span>b_H = {optimizedParams.b_h.toFixed(3)}</span>
149 |                 <span>n = {optimizedParams.n.toFixed(1)}</span>
150 |                 <span>α = {optimizedParams.alpha.toFixed(2)}</span>
151 |               </div>
152 |             </div>
153 |           )
154 |         </div>
155 |         {/* Severity Progress Bar */}
156 |         <div className="pt-4">
157 |           <div className="w-full h-2.5 bg-slate-100 rounded-full
158 |             overflow-hidden border border-slate-200">
159 |             <div
160 |               className={`h-full transition-all duration-500 ${
161 |                 pfvi >= 225
162 |                 ? 'bg-red-600'

```

```

121 |         : pfvi >= 150
122 |         ? 'bg-orange-500'
123 |         : pfvi >= 75
124 |         ? 'bg-amber-500'
125 |         : 'bg-emerald-600'
126 |     }`}`
127 |     style={{ width: `${pfviPercentage}%` }}
128 |   />
129 | </div>
130 | <div className="flex justify-between text-[10px] font-mono
font-semibold text-slate-500 mt-1.5">
131 |   <span>0 (Basah)</span>
132 |   <span>75 (Low)</span>
133 |   <span>150 (Mod)</span>
134 |   <span>225 (High)</span>
135 |   <span>300 (Extreme)</span>
136 | </div>
137 | </div>
138 | </div>
139 |
140 |   { /* Multi-Source Fire Intelligence Cockpit Bar */ }
141 |   <div className="telemetry-panel rounded-xl p-4 border
border-slate-200 bg-white shadow-sm grid grid-cols-1 md:grid-cols-3
gap-4">
142 |     { /* FirePing Proximity */ }
143 |     <div className="bg-slate-50 p-3.5 rounded-lg border
border-slate-200 flex items-center justify-between">
144 |       <div>
145 |         <span className="text-[10px] font-mono uppercase
tracking-wider font-bold text-slate-500 block mb-0.5">
146 |           FirePing Satellite Proximity
147 |         </span>
148 |         <div className="text-lg font-mono font-bold
text-slate-900">
149 |           {nearby?.nearest_distance_km !== null &&
nearby?.nearest_distance_km !== undefined
150 |             ? `${nearby?.nearest_distance_km} km`
151 |             : 'Aman (>25 km)'}
152 |         </div>
153 |         <span className="text-[10px] font-mono text-slate-500">
154 |           {nearby?.detection_count || 0} Deteksi (24j)
({nearby?.satellite_sources?.join(', ') || 'Modis/Viirs'})
155 |         </span>
156 |       </div>
157 |       <div className={`${px-2 py-1 text-[10px] font-mono font-bold
rounded border ${nearby?.detection_count ? 'bg-red-100 text-red-700
border-red-300' : 'bg-emerald-100 text-emerald-700
border-emerald-300'}}`}>
158 |         {nearby?.detection_count ? 'DETEKSI' : 'CLEAR'}
159 |       </div>
160 |     </div>
161 |
162 |     { /* GWIS Burned Area */ }
163 |     <div className="bg-slate-50 p-3.5 rounded-lg border
border-slate-200 flex items-center justify-between">
164 |       <div>
165 |         <span className="text-[10px] font-mono uppercase
tracking-wider font-bold text-slate-500 block mb-0.5">
166 |           GWIS 7-Day Burned Area
167 |         </span>
168 |         <div className="text-lg font-mono font-bold
text-amber-600">
169 |           {nearby?.burned_area_ha ?
nearby.burned_area_ha.toLocaleString('id-ID') : '0'} <span
className="text-xs text-slate-500 font-normal">Ha</span>
170 |         </div>
171 |         <span className="text-[10px] font-mono text-slate-500">
172 |           Estimasi Perimeter Satelit Global
173 |         </span>
174 |       </div>
175 |       <div className="px-2 py-1 text-[10px] font-mono font-bold
rounded border bg-amber-100 text-amber-800 border-amber-300">
176 |         GWIS NRT
177 |       </div>
178 |     </div>
179 |
180 |     { /* Weather FWI Index */ }
181 |     <div className="bg-slate-50 p-3.5 rounded-lg border
border-slate-200 flex items-center justify-between">
182 |       <div>
183 |         <span className="text-[10px] font-mono uppercase
tracking-wider font-bold text-slate-500 block mb-0.5">
184 |           Atmospheric Fire Weather (FWI)
185 |         </span>

```

```

186 |         <div className="text-lg font-mono font-bold
187 |             text-slate-900">
188 |             {fwi?.fwi_score !== undefined ?
189 |             fwi.fwi_score.toFixed(1) : '24.5'} <span className="text-xs
190 |             text-slate-500 font-normal">FWI</span>
191 |         </div>
192 |         <span className="text-[10px] font-mono text-slate-500">
193 |             {fwi?.source || 'OpenWeather / Canadian Engine'}
194 |         </span>
195 |     </div>
196 |     </div>
197 | </div>
198 |
199 |     {/* High-Density Environmental Telemetry Cockpit Bar */}
200 |     <div className="telemetry-panel bg-white border
201 |     border-slate-200 rounded-xl grid grid-cols-2 lg:grid-cols-4 divide-y
202 |     lg:divide-y-0 lg:divide-x divide-slate-200 shadow-sm">
203 |         {/* WT */}
204 |         <div className="p-4">
205 |             <span className="text-[11px] font-mono font-bold uppercase
206 |             tracking-wider text-slate-500 block mb-1">
207 |                 Water Table (WT)
208 |             </span>
209 |             <div className="text-xl font-mono font-bold
210 |             text-slate-900">
211 |                 {waterTable.toFixed(2)} <span className="text-xs
212 |                 font-normal text-slate-500">m</span>
213 |             </div>
214 |             <span className="text-[11px] text-slate-500 font-medium
215 |             mt-0.5 block">
216 |                 {waterTable < -0.8 ? 'Kritis Rendah' : 'Normal'}
217 |             </span>
218 |         </div>
219 |
220 |         {/* SM */}
221 |         <div className="p-4 lg:pl-6">
222 |             <span className="text-[11px] font-mono font-bold uppercase
223 |             tracking-wider text-slate-500 block mb-1">
224 |                 Soil Moisture (SM)
225 |             </span>
226 |             <div className="text-xl font-mono font-bold
227 |             text-slate-900">
228 |                 {soilMoisture.toFixed(1)} <span className="text-xs
229 |                 font-normal text-slate-500">%</span>
230 |             </div>
231 |             <span className="text-[11px] text-slate-500 font-medium
232 |             mt-0.5 block">
233 |                 {soilMoisture < 45 ? 'Tanah Kering' : 'Lembab'}
234 |             </span>
235 |         </div>
236 |
237 |         {/* Rf */}
238 |         <div className="p-4 lg:pl-6">
239 |             <span className="text-[11px] font-mono font-bold uppercase
240 |             tracking-wider text-slate-500 block mb-1">
241 |                 Rainfall (Rf)
242 |             </span>
243 |             <div className="text-xl font-mono font-bold
244 |             text-slate-900">
245 |                 {rainfall.toFixed(1)} <span className="text-xs
246 |                 font-normal text-slate-500">mm/d</span>
247 |             </div>
248 |             <span className="text-[11px] text-slate-500 font-medium
249 |             mt-0.5 block">
250 |                 {rainfall < 5.1 ? 'Tanpa Hujan Efektif' : 'Hujan
251 |                 Efektif'}
252 |             </span>
253 |         </div>
254 |
255 |         {/* Temp */}
256 |         <div className="p-4 lg:pl-6">
257 |             <span className="text-[11px] font-mono font-bold uppercase
258 |             tracking-wider text-slate-500 block mb-1">
259 |                 Temp Max (Temp)
260 |             </span>
261 |             <div className="text-xl font-mono font-bold
262 |             text-slate-900">
263 |                 {temp.toFixed(1)} <span className="text-xs font-normal
264 |                 text-slate-500">°C</span>
265 |             </div>

```

```

248 |         <span className="text-[11px] text-slate-500 font-medium
    |         mt-0.5 block">
249 |             {temp > 34.5 ? 'Suhu Tinggi' : 'Normal'}
250 |         </span>
251 |     </div>
252 | </div>
253 | </div>
254 | );
255 | };

```

Screenshot Aplikasi:

[Screenshot: Risk Status Gauge: PFVI, Fire Intelligence Cockpit, dan telemetri lingkungan — ss_gauge.png]

FITUR 3: Peta Satelit Live Interaktif (NASA FIRMS, VIIRS 375m, GFW Gambut, GIBS)

Deskripsi & Tujuan Fitur: Peta GIS interaktif berbasis **Leaflet** dengan empat overlay satelit live yang dapat di-toggle:

FIRMS WMS (NASA Key) — layer thermal VIIRS/MODIS 24 jam; **Hotspots API** — titik api vektor NASA FIRMS (VIIRS_SNPP_NRT) dengan jumlah hotspot hari ini; **GFW Vector Gambut** — titik api pada poligon lahan gambut Indonesia; **NASA GIBS 375m** — citra thermal anomalies VIIRS SNPP pass siang/malam. Marker lokasi stasiun berwarna sesuai risiko PFVI (hijau → merah). Pengguna dapat **klik titik mana pun di peta** untuk membuat target inspeksi baru, mendapatkan koordinat & jarak ke stasiun utama; aplikasi otomatis memuat ulang pipeline untuk koordinat tersebut (fly-to animasi).

Cara Menggunakan:

- Gunakan peta untuk melihat sebaran titik api di sekitar lokasi terpilih (marker oranye = FIRMS, merah = GFW).
- Klik tombol toggle di pojok kanan atas peta untuk mengaktifkan/menonaktifkan tiap lapisan satelit (jumlah hotspot tampil pada tombol **Hotspots API**).
- Klik sembarang titik pada peta — sebuah marker target biru muncul dan panel inspeksi menampilkan latitude/longitude & jarak ke stasiun; pipeline otomatis dijalankan untuk titik tersebut.
- Klik marker hotspot untuk melihat detail (brightness TI4, FRP, tanggal akuisisi, satelit, confidence, day/night).

Kode Sumber (Source Code Listing): frontend/src/components/PeatlandMap.tsx

```

1 | import React, { useEffect, useState } from 'react';
2 | import { MapContainer, TileLayer, WMSTileLayer, Marker, Popup,
    |   Circle, useMap, useMapEvents } from 'react-leaflet';
3 | import L from 'leaflet';
4 | import { MapPin, Flame, Satellite, ShieldAlert, Radio, Crosshair,
    |   Zap, Navigation } from 'lucide-react';
5 | import axios from 'axios';
6 |
7 | export interface LocationPreset {
8 |   id: string;
9 |   name: string;
10 |   lat: number;
11 |   lon: number;
12 | }
13 |
14 | export interface GfwPeatlandFire {
15 |   latitude: number;
16 |   longitude: number;
17 |   alert_date: string;
18 |   confidence_cat?: string;
19 |   bright_ti4_K?: number;
20 |   frp_MW?: number;
21 |   is_peatland?: boolean;
22 |   adm1?: string;
23 |   adm2?: string;
24 | }
25 |
26 | export interface FirmsHotspot {
27 |   latitude: number;
28 |   longitude: number;
29 |   bright_ti4: number;
30 |   frp: number;

```

```

31 |   acq_date: string;
32 |   acq_time: string;
33 |   satellite: string;
34 |   instrument: string;
35 |   confidence: string;
36 |   daynight: string;
37 | }
38 |
39 | const FIRMS_MAP_KEY = "aa16407e5eb11df46b09cafc085fe020";
40 |
41 | function getDistanceKm(lat1: number, lon1: number, lat2: number,
42 |   lon2: number): number {
43 |   const R = 6371;
44 |   const dLat = (lat2 - lat1) * (Math.PI / 180);
45 |   const dLon = (lon2 - lon1) * (Math.PI / 180);
46 |   const a =
47 |     Math.sin(dLat / 2) * Math.sin(dLat / 2) +
48 |     Math.cos(lat1 * (Math.PI / 180)) * Math.cos(lat2 * (Math.PI /
49 |       180)) *
50 |     Math.sin(dLon / 2) * Math.sin(dLon / 2);
51 |   const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
52 |   return R * c;
53 | }
54 |
55 | const createCustomIcon = (color: string) => {
56 |   return L.divIcon({
57 |     className: 'custom-leaflet-marker',
58 |     html: `<div style="background-color: ${color}; width: 18px;
59 |       height: 18px; border-radius: 50%; border: 2.5px solid white;
60 |       box-shadow: 0 2px 10px rgba(0,0,0,0.5); cursor: pointer;"></div>`,
61 |     iconSize: [18, 18],
62 |     iconAnchor: [9, 9]
63 |   });
64 | };
65 |
66 | const createPeatlandFireIcon = () => {
67 |   return L.divIcon({
68 |     className: 'gfw-peatland-fire-marker',
69 |     html: `<div style="background-color: #dc2626; width: 14px;
70 |       height: 14px; border-radius: 50%; border: 2px solid #fee2e2;
71 |       box-shadow: 0 0 12px #dc2626; animation: pulse 1.5s infinite;
72 |       cursor: pointer;"></div>`,
73 |     iconSize: [14, 14],
74 |     iconAnchor: [7, 7]
75 |   });
76 | };
77 |
78 | const createFirmsHotspotIcon = () => {
79 |   return L.divIcon({
80 |     className: 'firms-hotspot-marker',
81 |     html: `<div style="background-color: #ea580c; width: 12px;
82 |       height: 12px; border-radius: 50%; border: 2px solid #ffedd5;
83 |       box-shadow: 0 0 10px #ea580c; cursor: pointer;"></div>`,
84 |     iconSize: [12, 12],
85 |     iconAnchor: [6, 6]
86 |   });
87 | };
88 |
89 | const createTargetIcon = () => {
90 |   return L.divIcon({
91 |     className: 'clicked-target-marker',
92 |     html: `<div style="background-color: #0284c7; width: 16px;
93 |       height: 16px; border-radius: 50%; border: 2.5px solid white;
94 |       box-shadow: 0 0 12px #0284c7; cursor: pointer;"></div>`,
95 |     iconSize: [16, 16],
96 |     iconAnchor: [8, 8]
97 |   });
98 | };
99 |
100 | function MapFlyTo({ lat, lon }: { lat: number; lon: number }) {
101 |   const map = useMap();
102 |   useEffect(() => {
103 |     map.flyTo([lat, lon], 10, { duration: 1.5 });
104 |   }, [lat, lon, map]);
105 |   return null;
106 | }
107 |
108 | function MapClickHandler({ onMapClick }: { onMapClick: (lat: number,
109 |   lon: number) => void }) {
110 |   useMapEvents({
111 |     click(e) {
112 |       onMapClick(e.latlng.lat, e.latlng.lng);
113 |     }
114 |   });
115 | }

```

```

103 |   return null;
104 | }
105 |
106 | interface PeatlandMapProps {
107 |   currentPfvi: number;
108 |   status: string;
109 |   location: LocationPreset;
110 |   onSelectCustomLocation?: (lat: number, lon: number, name?: string)
    => void;
111 | }
112 |
113 | export const PeatlandMap: React.FC<PeatlandMapProps> = ({
114 |   currentPfvi,
115 |   status,
116 |   location,
117 |   onSelectCustomLocation
118 | }) => {
119 |   const pos: [number, number] = [location.lat, location.lon];
120 |
121 |   // Live Satellite Overlay Layers state
122 |   const [showFirmsWms, setShowFirmsWms] = useState(true);
123 |   const [showFirmsVector, setShowFirmsVector] = useState(true);
124 |   const [showGibsViirsDay, setShowGibsViirsDay] = useState(true);
125 |   const [showGfwPeatland, setShowGfwPeatland] = useState(true);
126 |
127 |   // Interactivity state
128 |   const [clickedTarget, setClickedTarget] = useState<{ lat: number;
    lon: number } | null>(null);
129 |   const [selectedEntity, setSelectedEntity] = useState<{
130 |     type: 'hotspot' | 'gfw' | 'station' | 'point';
131 |     title: string;
132 |     lat: number;
133 |     lon: number;
134 |     details: Record<string, any>;
135 |   } | null>(null);
136 |
137 |   const [gfwFires, setGfwFires] = useState<GfwPeatlandFire[]>([]);
138 |   const [firmsHotspots, setFirmsHotspots] =
    useState<FirmsHotspot[]>([]);
139 |   const [severeAlerts, setSevereAlerts] = useState<any[]>([]);
140 |
141 |   useEffect(() => {
142 |     axios.get('/api/v1/fire-intelligence/severe-alerts')
143 |       .then((res) => {
144 |         if (res.data && res.data.alerts) {
145 |           setSevereAlerts(res.data.alerts);
146 |         }
147 |       })
148 |       .catch((err) => console.warn('Could not fetch severe fire
    alerts:', err));
149 |   }, []);
150 |
151 |   useEffect(() => {
152 |     if (showGfwPeatland) {
153 |       axios.get('/api/v1/fire-intelligence/gfw-peatland-fires?limit=100')
154 |         .then((res) => {
155 |           if (res.data && res.data.fires) {
156 |             setGfwFires(res.data.fires);
157 |           }
158 |         })
159 |         .catch((err) => console.warn('Could not fetch GFW peatland
    fires:', err));
160 |     }
161 |   }, [showGfwPeatland]);
162 |
163 |   useEffect(() => {
164 |     if (showFirmsVector) {
165 |       axios.get('/api/v1/fire-intelligence/firms-hotspots?day_range=1')
166 |         .then((res) => {
167 |           if (res.data && res.data.hotspots) {
168 |             setFirmsHotspots(res.data.hotspots);
169 |           }
170 |         })
171 |         .catch((err) => console.warn('Could not fetch NASA FIRMS
    vector hotspots:', err));
172 |     }
173 |   }, [showFirmsVector]);
174 |
175 |   const handleMapClick = (lat: number, lon: number) => {
176 |     setClickedTarget({ lat, lon });
177 |     const distKm = getDistanceKm(location.lat, location.lon, lat,
    lon);
178 |     setSelectedEntity({
179 |       type: 'point',

```

```

180 |     title: `Koordinat Klik Peta Satelit`,
181 |     lat,
182 |     lon,
183 |     details: {
184 |       'Latitude': lat.toFixed(5),
185 |       'Longitude': lon.toFixed(5),
186 |       'Jarak ke Stasiun Utama': `${distKm.toFixed(2)} km`,
187 |       'Status Wilayah': 'Realtime Satellite Grid Pick'
188 |     }
189 |   });
190 | };
191 |
192 | const getMarkerColor = (pfvi: number) => {
193 |   if (pfvi >= 225) return '#dc2626'; // Red
194 |   if (pfvi >= 150) return '#ea580c'; // Orange
195 |   if (pfvi >= 75) return '#d97706'; // Amber
196 |   return '#16a34a'; // Green
197 | };
198 |
199 | const color = getMarkerColor(currentPfvi);
200 |
201 | return (
202 |   <div className="telemetry-panel rounded-xl p-5 mb-6 bg-white
border border-slate-200 shadow-sm">
203 |     <div className="flex flex-col lg:flex-row items-start
lg:items-center justify-between gap-3 mb-4">
204 |       <div className="flex items-center gap-2">
205 |         <Satellite className="w-5 h-5 text-emerald-600 shrink-0"
/>
206 |         <div>
207 |           <h3 className="text-sm font-bold tracking-wide
text-slate-900 flex items-center gap-2 flex-wrap">
208 |             REAL LIVE SATELLITE FIRE MAP (NASA FIRMS & VIIRS
OVERPASS)
209 |             <span className="inline-flex items-center gap-1 px-2
py-0.5 text-[10px] font-mono bg-emerald-100 text-emerald-800 border
border-emerald-300 rounded font-semibold">
210 |               <Radio className="w-3 h-3 text-emerald-600
animate-pulse" />
211 |               NASA FIRMS MAP_KEY ACTIVE
212 |             </span>
213 |           </h3>
214 |           <p className="text-xs text-slate-500 font-medium">
Peta Satelit Live Resmi NASA FIRMS, VIIRS 375m & GFW
215 |           Gambut. Klik lokasi manapun untuk inspeksi titik api.
216 |         </p>
217 |       </div>
218 |     </div>
219 |
220 |     { /* Live Satellite Layer Toggles */ }
221 |     <div className="flex flex-wrap items-center gap-2 text-xs
font-mono">
222 |       <button
223 |         onClick={() => setShowFirmsWms(!showFirmsWms)}
224 |         className={`px-3 py-1.5 rounded border transition-colors
flex items-center gap-1.5 font-bold ${
225 |           showFirmsWms
226 |             ? 'bg-rose-600 text-white border-rose-700 shadow-xs'
227 |             : 'bg-slate-100 text-slate-600 border-slate-300'
228 |         }`>
229 |         >
230 |         <Flame className="w-3.5 h-3.5" />
231 |         <span>FIRMS WMS (NASA Key)</span>
232 |       </button>
233 |
234 |       <button
235 |         onClick={() => setShowFirmsVector(!showFirmsVector)}
236 |         className={`px-3 py-1.5 rounded border transition-colors
flex items-center gap-1.5 font-bold ${
237 |           showFirmsVector
238 |             ? 'bg-orange-600 text-white border-orange-700
shadow-xs'
239 |             : 'bg-slate-100 text-slate-600 border-slate-300'
240 |         }`>
241 |         >
242 |         <Radio className="w-3.5 h-3.5" />
243 |         <span>Hotspots API ({firmsHotspots.length})</span>
244 |       </button>
245 |
246 |       <button
247 |         onClick={() => setShowGfwPeatland(!showGfwPeatland)}
248 |         className={`px-3 py-1.5 rounded border transition-colors
flex items-center gap-1.5 font-bold ${
249 |           showGfwPeatland

```

```

250 |         ? 'bg-red-600 text-white border-red-700 shadow-xs'
251 |         : 'bg-slate-100 text-slate-600 border-slate-300'
252 |     }` }
253 | >
254 |     <ShieldAlert className="w-3.5 h-3.5" />
255 |     <span>GFW Vector Gambut</span>
256 | </button>
257 |
258 |     <button
259 |       onClick={() => setShowGibsViirsDay(!showGibsViirsDay)}
260 |       className={`px-3 py-1.5 rounded border transition-colors
flex items-center gap-1.5 font-bold ${
261 |         showGibsViirsDay
262 |           ? 'bg-cyan-700 text-white border-cyan-800 shadow-xs'
263 |           : 'bg-slate-100 text-slate-600 border-slate-300'
264 |         }` }
265 |     >
266 |       <Satellite className="w-3.5 h-3.5" />
267 |       <span>NASA GIBS 375m</span>
268 |     </button>
269 |   </div>
270 | </div>
271 |
272 |   { /* Severe Fire Alerts Ticker */ }
273 |   { severeAlerts.length > 0 && (
274 |     <div className="mb-4 bg-red-50 border border-red-200
rounded-lg p-3 text-xs font-mono shadow-sm">
275 |       <div className="flex items-center justify-between mb-2">
276 |         <span className="font-bold text-red-700 flex
items-center gap-1.5 uppercase tracking-wide">
277 |           <Flame className="w-4 h-4 text-red-600 animate-pulse"
/>
278 |           PERINGATAN KEBAKARAN PARAH (SEVERE FIRE ALERTS)
279 |         </span>
280 |         <span className="text-[10px] text-red-800 bg-red-100
border border-red-300 px-2 py-0.5 rounded font-bold">
281 |           {severeAlerts.length} Area Terdampak Terdeteksi
282 |         </span>
283 |       </div>
284 |       <div className="grid grid-cols-1 sm:grid-cols-2
lg:grid-cols-4 gap-2">
285 |         {severeAlerts.map((alert) => (
286 |           <button
287 |             key={alert.id}
288 |             onClick={() => {
289 |               setSelectedEntity({
290 |                 type: 'gfw',
291 |                 title: alert.title,
292 |                 lat: alert.lat,
293 |                 lon: alert.lon,
294 |                 details: {
295 |                   'Lokasi Kebakaran': alert.location,
296 |                   'Estimasi Luas Terbakar':
` ${alert.estimated_burned_km2} km² (${alert.estimated_burned_ha}
Ha)`,
297 |                   'Intensitas FRP Maks': ` ${alert.frp_max_mw}
MW`,
298 |                   'Satelit Pengawas': alert.satellite_sensor,
299 |                   'Tingkat Bahaya': alert.severity,
300 |                   'Status Waktu': alert.updated_ago
301 |                 }
302 |               });
303 |               if (onSelectCustomLocation) {
304 |                 onSelectCustomLocation(alert.lat, alert.lon,
alert.location);
305 |               }
306 |             }}
307 |             className="text-left bg-white hover:bg-red-50 p-2.5
rounded border border-red-200 hover:border-red-400 transition-all
group shadow-xs cursor-pointer"
308 |           >
309 |             <div className="font-bold text-red-700 truncate flex
items-center justify-between">
310 |               <span className="group-hover:text-red-900
transition-colors">{alert.title}</span>
311 |               <span className="text-[10px] px-1.5 py-0.2
bg-red-100 text-red-800 rounded
font-bold">{alert.estimated_burned_km2} km²</span>
312 |             </div>
313 |             <div className="text-[11px] text-slate-700 truncate
mt-1 font-medium">{alert.location}</div>
314 |             <div className="text-[10px] text-slate-500 mt-1.5
flex items-center justify-between pt-1 border-t border-slate-100">
315 |               <span>{alert.updated_ago}</span>

```

```

316 |         <span className="text-emerald-700 font-bold
317 | group-hover:underline">Fokus Peta →</span>
318 |     </div>
319 |     </button>
320 |   )))
321 | </div>
322 | })
323 |
324 | {/* Dedicated Real Live Satellite Map Display */}
325 | <div className="grid grid-cols-1 lg:grid-cols-4 gap-4">
326 |   <div className="lg:col-span-3 h-[460px] w-full rounded-lg
327 | overflow-hidden relative border border-slate-300 shadow-sm">
328 |     <MapContainer
329 |       center={pos}
330 |       zoom={10}
331 |       scrollWheelZoom={true}
332 |       className="h-full w-full cursor-crosshair"
333 |     >
334 |       <MapFlyTo lat={location.lat} lon={location.lon} />
335 |       <MapClickHandler onMapClick={handleMapClick} />
336 |       {/* High-Resolution Satellite Base Layer (Esri World
337 | Imagery) */}
338 |       <TileLayer
339 |         attribution='&copy; <a
340 | href="https://www.esri.com/">Esri World Imagery</a>'
341 |         url="https://server.arcgisonline.com/ArcGIS/rest/servi
342 | ces/World_Imagery/MapServer/tile/{z}/{y}/{x}"
343 |       />
344 |       {/* Transparent Administrative Place Names & Boundaries
345 | Label Overlay */}
346 |       <TileLayer
347 |         attribution='&copy; <a href="https://carto.com/">CARTO
348 | Voyager Labels</a>'
349 |         url="https://{s}.basemaps.cartocdn.com/rastertiles/voy
350 | ager_only_labels/{z}/{x}/{y}/{r}.png"
351 |       />
352 |       {/* NASA FIRMS Dedicated Authorized WMS Thermal Layer
353 | (MAP_KEY Authorized) */}
354 |       {showFirmsWms && (
355 |         <WMSTileLayer
356 |           url={`https://firms.modaps.eosdis.nasa.gov/mapserver
357 | /wms/fires/${FIRMS_MAP_KEY}/`}
358 |           layers="fires_viirs_24, fires_modis_24"
359 |           format="image/png"
360 |           transparent={true}
361 |           attribution="NASA FIRMS Dedicated WMS (MAP_KEY
362 | Active)"
363 |         />
364 |       )}
365 |       {/* NASA GIBS VIIRS 375m Thermal Anomalies Web Mercator
366 | (EPSG3857) WMS Layer */}
367 |       {showGibsViirsDay && (
368 |         <WMSTileLayer
369 |           url="https://gibs.earthdata.nasa.gov/wms/epsg3857/best/wms.cgi"
370 |           layers="VIIRS_SNPP_Thermal_Anomalies_375m_Day, VIIRS_
371 | SNPP_Thermal_Anomalies_375m_Night"
372 |           format="image/png"
373 |           transparent={true}
374 |           attribution="NASA GIBS VIIRS 375m Thermal"
375 |         />
376 |       )}
377 |       {/* NASA FIRMS Vector Hotspot Point Markers */}
378 |       {showFirmsVector && firmsHotspots.map((hotspot, idx) =>
379 |         (
380 |           <Marker
381 |             key={`firms-vector-${idx}`}
382 |             position={[hotspot.latitude, hotspot.longitude]}
383 |             icon={createFirmsHotspotIcon()}
384 |             eventHandlers={{
385 |               click: (e) => {
386 |                 e.originalEvent.stopPropagation();
387 |                 const distKm = getDistanceKm(location.lat,
388 | location.lon, hotspot.latitude, hotspot.longitude);
389 |                 setSelectedEntity({
390 |                   type: 'hotspot',
391 |                   title: `NASA FIRMS Hotspot (VIIRS)`,
392 |                   lat: hotspot.latitude,
393 |                   lon: hotspot.longitude,

```

```

385 |         details: {
386 |             'Tanggal / Jam UTC': `${hotspot.acq_date}
387 |             'Kecerahan (T_i4)': `${hotspot.bright_ti4}
388 |             'Daya FRP': hotspot.frp > 0 ?
389 |             'Satelit Sensor': `${hotspot.satellite}
390 |             'Tingkat Kepercayaan': hotspot.confidence,
391 |             'Jarak ke Stasiun': `${distKm.toFixed(2)}
392 |         }
393 |     });
394 | }
395 | }}
396 | >
397 |     <Popup className="custom-popup">
398 |         <div className="p-2 font-mono text-xs
399 |         text-slate-900">
400 |             <div className="font-bold text-orange-600 mb-1
401 |             flex items-center gap-1">
402 |                 <Flame className="w-3.5 h-3.5 inline" /> NASA
403 |                 FIRMS Hotspot (VIIRS)
404 |                 </div>
405 |                 <div>Tgl / Jam: {hotspot.acq_date}
406 |                 {hotspot.acq_time} UTC</div>
407 |                 <div>Kecerahan (T_i4): {hotspot.bright_ti4}
408 |                 K</div>
409 |                 <div>Daya FRP: {hotspot.frp > 0 ?
410 |                 `${hotspot.frp} MW` : 'N/A'</div>
411 |                 <div>Satelit: {hotspot.satellite}
412 |                 ({hotspot.instrument})</div>
413 |                 <div className="mt-1 text-[10px] text-orange-700
414 |                 font-bold">✓ Authorized NASA FIRMS Key</div>
415 |                 </div>
416 |             </Popup>
417 |         </Marker>
418 |     )}}
419 |
420 |     {/* GFW Vector Active Fires on Peatlands */}
421 |     {showGfwPeatland && gfwFires.map((fire, idx) => (
422 |         <Marker
423 |             key={`gfw-fire-${idx}`}
424 |             position={[fire.latitude, fire.longitude]}
425 |             icon={createPeatlandFireIcon()}
426 |             eventHandlers={{
427 |                 click: (e) => {
428 |                     e.originalEvent.stopPropagation();
429 |                     const distKm = getDistanceKm(location.lat,
430 |                     location.lon, fire.latitude, fire.longitude);
431 |                     setSelectedEntity({
432 |                         type: 'gfw',
433 |                         title: `GFW Vector Fire on Peatland`,
434 |                         lat: fire.latitude,
435 |                         lon: fire.longitude,
436 |                         details: {
437 |                             'Tanggal Alert': fire.alert__date,
438 |                             'Kabupaten / Provinsi': `${fire.adm2 ||
439 |                             'Kab'}`, ${fire.adm1 || 'Prov'}`,
440 |                             'Kecerahan (K)': fire.bright_ti4_K ?
441 |                             `${fire.bright_ti4_K} K` : '-',
442 |                             'Daya FRP': fire.frp_MW ? `${fire.frp_MW}
443 |                             MW` : '-',
444 |                             'Indikator Gambut': 'is__peatland = true',
445 |                             'Jarak ke Stasiun': `${distKm.toFixed(2)}
446 |                             km`
447 |                         }
448 |                     });
449 |                 }}
450 |             }}
451 |         >
452 |             <Popup className="custom-popup">
453 |                 <div className="p-2 font-mono text-xs
454 |                 text-slate-900">
455 |                     <div className="font-bold text-red-600 mb-1 flex
456 |                     items-center gap-1">
457 |                         <Flame className="w-3.5 h-3.5 inline" />
458 |                         Active Fire on Peatland
459 |                     </div>
460 |                     <div>Tgl Alert: {fire.alert__date}</div>
461 |                     <div>Lokasi: {fire.adm2 || 'Kab'}`, {fire.adm1 ||
462 |                     'Prov'}</div>
463 |                     <div>Kecerahan: {fire.bright_ti4_K || '-'}

```

```

K</div>
447 |         <div>Daya FRP: {fire.frp__MW ? `${fire.frp__MW}
MW` : '-'}</div>
448 |         <div className="mt-1 text-[10px]
text-emerald-700 font-bold">✓ Verified GFW Peatland Layer</div>
449 |       </div>
450 |     </Popup>
451 |   </Marker>
452 | ))}
453 |
454 |   {/* User Clicked Target Point Marker */}
455 |   {clickedTarget && (
456 |     <Marker position={{[clickedTarget.lat,
clickedTarget.lon]} icon={createTargetIcon()}>
457 |       <Popup className="custom-popup">
458 |         <div className="p-2 font-mono text-xs
text-slate-900">
459 |           <div className="font-bold text-sky-600 mb-1 flex
items-center gap-1">
460 |             <Crosshair className="w-3.5 h-3.5 inline" />
Titik Peta Satelit Terpilih
461 |           </div>
462 |           <div>Lat: {clickedTarget.lat.toFixed(5)}</div>
463 |           <div>Lon: {clickedTarget.lon.toFixed(5)}</div>
464 |           <button
465 |             onClick={() => onSelectCustomLocation &&
onSelectCustomLocation(clickedTarget.lat, clickedTarget.lon)}
466 |             className="mt-2 w-full px-2 py-1
bg-emerald-600 hover:bg-emerald-700 text-white font-semibold rounded
text-[11px] transition-colors cursor-pointer"
467 |             >
468 |               Analisis Telemetry Titik Ini
469 |             </button>
470 |           </div>
471 |         </Popup>
472 |       </Marker>
473 |     )}
474 |
475 |   {/* Telemetry Station Marker */}
476 |   <Marker
477 |     position={pos}
478 |     icon={createCustomIcon(color)}
479 |     eventHandlers={{
480 |       click: (e) => {
481 |         e.originalEvent.stopPropagation();
482 |         setSelectedEntity({
483 |           type: 'station',
484 |           title: `Stasiun Telemetry Utama`,
485 |           lat: location.lat,
486 |           lon: location.lon,
487 |           details: {
488 |             'Nama Lokasi': location.name,
489 |             'PFVI Score': `${currentPfvi.toFixed(1)} /
300.0`,
490 |             'Status Risiko': status,
491 |             'Koordinat': `${location.lat},
${location.lon}`
492 |           }
493 |         });
494 |       }
495 |     }}
496 |   >
497 |     <Popup className="custom-popup">
498 |       <div className="p-2 text-slate-900">
499 |         <h4 className="font-bold text-slate-900 text-sm
mb-1">{location.name}</h4>
500 |         <p className="text-xs text-slate-600">Stasiun
Telemetry & Multi-Satelit Overpass</p>
501 |         <div className="mt-2 text-xs font-mono
font-semibold text-slate-900">
502 |           PFVI Score: {currentPfvi.toFixed(1)} / 300.0
({status})
503 |         </div>
504 |       </div>
505 |     </Popup>
506 |   </Marker>
507 |
508 |   <Circle
509 |     center={pos}
510 |     radius={currentPfvi >= 225 ? 15000 : 8000}
511 |     pathOptions={{
512 |       color: color,
513 |       fillColor: color,
514 |       fillOpacity: 0.15,

```

```

515 |         weight: 1.5
516 |     }}
517 | }
518 | </MapContainer>
519 |
520 | { /* Legend Overlay */
521 |   <div className="absolute bottom-3 left-3 z-[1000]
bg-white/95 backdrop-blur-xs p-3 rounded-lg border border-slate-300
text-xs shadow-md space-y-2 max-w-[90%] text-slate-800">
522 |     <div className="flex items-center gap-3 font-mono
flex-wrap font-semibold text-[11px]">
523 |       <div className="flex items-center gap-1.5">
524 |         <span className="w-2.5 h-2.5 rounded-full
bg-emerald-600 inline-block" />
525 |         <span>Low (&lt;75)</span>
526 |       </div>
527 |       <div className="flex items-center gap-1.5">
528 |         <span className="w-2.5 h-2.5 rounded-full
bg-amber-500 inline-block" />
529 |         <span>Moderate</span>
530 |       </div>
531 |       <div className="flex items-center gap-1.5">
532 |         <span className="w-2.5 h-2.5 rounded-full
bg-orange-500 inline-block" />
533 |         <span>High</span>
534 |       </div>
535 |       <div className="flex items-center gap-1.5">
536 |         <span className="w-2.5 h-2.5 rounded-full bg-red-600
inline-block" />
537 |         <span>Extreme (&gt;225)</span>
538 |       </div>
539 |       <div className="flex items-center gap-1.5 border-l
border-slate-300 pl-2">
540 |         <span className="w-2.5 h-2.5 rounded-full
bg-orange-600 inline-block" />
541 |         <span className="text-orange-700 font-bold">FIRMS
Hotspot</span>
542 |       </div>
543 |       <div className="flex items-center gap-1.5 pl-1">
544 |         <span className="w-2.5 h-2.5 rounded-full bg-red-600
animate-ping inline-block" />
545 |         <span className="text-red-700 font-bold">GFW
Gambut</span>
546 |       </div>
547 |     </div>
548 |   </div>
549 | }
550 |
551 | { /* Dynamic Satellite & Entity Inspector Panel */
552 |   <div className="bg-slate-50 border border-slate-200
rounded-lg p-4 flex flex-col justify-between shadow-xs">
553 |     <div>
554 |       <div className="flex items-center justify-between
border-b border-slate-200 pb-3 mb-3">
555 |         <h4 className="text-xs font-bold tracking-wider
text-emerald-700 uppercase font-mono flex items-center gap-1.5">
556 |           <Navigation className="w-3.5 h-3.5 text-emerald-600"
/>
557 |           SATELLITE HOTSPOT INSPECTOR
558 |         </h4>
559 |         <span className="text-[10px] font-mono px-2 py-0.5
bg-slate-200 rounded text-slate-700 font-bold">
560 |           {selectedEntity ? selectedEntity.type.toUpperCase()
: 'TELEMETRI'}
561 |         </span>
562 |       </div>
563 |
564 |       {selectedEntity ? (
565 |         <div className="space-y-3 font-mono text-xs">
566 |           <div className="text-slate-900 font-bold flex
items-center gap-1.5">
567 |             {selectedEntity.type === 'hotspot' && <Flame
className="w-4 h-4 text-orange-600 shrink-0" />}
568 |             {selectedEntity.type === 'gfw' && <ShieldAlert
className="w-4 h-4 text-red-600 shrink-0" />}
569 |             {selectedEntity.type === 'point' && <Crosshair
className="w-4 h-4 text-sky-600 shrink-0" />}
570 |             {selectedEntity.type === 'station' && <MapPin
className="w-4 h-4 text-emerald-600 shrink-0" />}
571 |             <span>{selectedEntity.title}</span>
572 |           </div>
573 |
574 |           <div className="bg-white p-3 rounded border
border-slate-200 space-y-2 text-slate-800 shadow-xs">

```

```

575 |         {Object.entries(selectedEntity.details).map(([k,
576 |           v]) => (
577 |             <div key={k} className="flex justify-between
578 |             items-center text-[11px]">
579 |               <span className="text-slate-500
580 |               font-medium">{k}</span>
581 |               <span className="font-bold
582 |               text-slate-900">{String(v)}</span>
583 |             </div>
584 |           )))
585 |       </div>
586 |       {onSelectCustomLocation && (
587 |         <button
588 |           onClick={() =>
589 |             onSelectCustomLocation(selectedEntity.lat, selectedEntity.lon,
590 |             selectedEntity.title)}
591 |           className="w-full mt-2 py-2 px-3 bg-emerald-600
592 |           hover:bg-emerald-700 text-white font-bold rounded text-xs
593 |           transition-colors flex items-center justify-center gap-2 shadow-xs
594 |           cursor-pointer"
595 |         >
596 |           <Zap className="w-3.5 h-3.5" />
597 |           <span>Fokuskan Telemetry ke Titik Ini</span>
598 |         </button>
599 |       )}
600 |     </div>
601 |   ) : (
602 |     <div className="py-8 text-center text-slate-500
603 |     font-mono text-xs space-y-3">
604 |       <Crosshair className="w-8 h-8 text-slate-400 mx-auto
605 |       animate-pulse" />
606 |       <p>Klik titik mana saja di peta satelit atau marker
607 |       hotspot kebakaran untuk inspeksi detail telemetry.</p>
608 |       <div className="text-[10px] text-emerald-800
609 |       bg-emerald-100 p-2 rounded border border-emerald-300 font-semibold">
610 |         ✓ High-Res Live Satellite Map & NASA FIRMS Key
611 |       </div>
612 |     </div>
613 |   );
614 | };

```

Screenshot Aplikasi:

[Screenshot: Peta Satelit Live: overlay FIRMS WMS, hotspots API, GFW gambut, dan GIBS 375m — ss_map.png]

FITUR 4: Kontrol Algoritma Pipeline (Imputasi, Model Forecasting, Horizon) — Mode Pakar

Deskripsi & Tujuan Fitur: Panel konfigurasi pipeline eksperimen untuk peneliti: **Imputasi Data Missing** (kNN Gower Distance, Cubic Spline, LOESS Smoothing, Linear Interpolation), **Model Time Series Forecasting** (ARIMA + Box-Cox stokastik, LSTM PyTorch, GRU PyTorch), dan **Horizon Proyeksi** (4 hari default jurnal, 7 hari, 14 hari). Tombol **Jalankan Pipeline Auto** memicu backend POST /api/v1/pipeline/auto yang menjalankan imputasi → forecasting → optimasi Nelder-Mead → simulasi PFVI → klasifikasi risiko secara berantai.

Cara Menggunakan:

1. Pilih metode imputasi (default: kNN) — digunakan bila data telemetry memiliki nilai kosong (NaN).
1. Pilih model forecasting (default: ARIMA; LSTM/GRU melatih jaringan saraf di backend).

1. Pilih horizon proyeksi (4/7/14 hari).

1. Tekan **Jalankan Pipeline Auto** dan tunggu hasil (indikator spinner saat proses berjalan); hasil langsung memperbarui gauge, grafik, dan peta.

Kode Sumber (Source Code Listing): frontend/src/components/PipelineControls.tsx, backend/main.py

frontend/src/components/PipelineControls.tsx:

```

1 | import React from 'react';
2 | import { Settings, Play, Database, Cpu, Calendar } from
   | 'lucide-react';
3 |
4 | interface PipelineControlsProps {
5 |   imputation: string;
6 |   setImputation: (val: string) => void;
7 |   model: string;
8 |   setModel: (val: string) => void;
9 |   h: number;
10 |   setH: (val: number) => void;
11 |   epochs: number;
12 |   setEpochs: (val: number) => void;
13 |   onRunPipeline: () => void;
14 |   isRunning: boolean;
15 | }
16 |
17 | export const PipelineControls: React.FC<PipelineControlsProps> = ({
18 |   imputation,
19 |   setImputation,
20 |   model,
21 |   setModel,
22 |   h,
23 |   setH,
24 |   epochs,
25 |   setEpochs,
26 |   onRunPipeline,
27 |   isRunning
28 | }) => {
29 |   return (
30 |     <div className="telemetry-panel bg-white border border-slate-200
   | rounded-xl p-5 mb-6 shadow-sm">
31 |       <div className="flex items-center justify-between mb-4
   | border-b border-slate-200 pb-3">
32 |         <div className="flex items-center gap-2">
33 |           <Settings className="w-4 h-4 text-emerald-600" />
34 |           <h3 className="text-sm font-bold text-slate-900 uppercase
   | tracking-wider">
35 |             KONFIGURASI ALGORITMA & MODEL PIPELINE
36 |           </h3>
37 |         </div>
38 |         <span className="text-xs text-slate-500 font-mono
   | font-semibold">FastAPI Engine</span>
39 |       </div>
40 |
41 |       <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4
   | gap-4 items-end">
42 |         </* Imputation Method Selection */>
43 |         <div>
44 |           <label className="block text-xs font-semibold
   | text-slate-700 mb-1.5 flex items-center gap-1.5">
45 |             <Database className="w-3.5 h-3.5 text-cyan-600" />
46 |             <span>Imputasi Data Missing:</span>
47 |           </label>
48 |           <select
   | value={imputation}
49 |           onChange={(e) => setImputation(e.target.value)}
50 |           className="w-full bg-slate-50 border border-slate-300
   | rounded-lg px-3 py-2 text-xs font-mono text-slate-800
51 |           focus:outline-none focus:border-emerald-600 cursor-pointer
   | font-medium"
52 |           >
53 |             <option value="knn">kNN (Gower Distance)</option>
54 |             <option value="spline">Cubic Spline
   | Interpolation</option>
55 |             <option value="loess">LOESS Smoothing</option>
56 |             <option value="linear">Linear Interpolation</option>
57 |           </select>
58 |         </div>

```

```

59 |
60 |     { /* Model Selection */ }
61 |     <div>
62 |         <label className="block text-xs font-semibold
63 | text-slate-700 mb-1.5 flex items-center gap-1.5">
64 |             <Cpu className="w-3.5 h-3.5 text-blue-600" />
65 |             <span>Model Time Series Forecasting:</span>
66 |         </label>
67 |         <select
68 |             value={model}
69 |             onChange={(e) => setModel(e.target.value)}
70 |             className="w-full bg-slate-50 border border-slate-300
71 | rounded-lg px-3 py-2 text-xs font-mono text-slate-800
72 | focus:outline-none focus:border-emerald-600 cursor-pointer
73 | font-medium"
74 |         >
75 |             <option value="arima">ARIMA + Box-Cox
76 | (Stokastik)</option>
77 |             <option value="lstm">LSTM Neural Network
78 | (PyTorch)</option>
79 |             <option value="gru">GRU Neural Network
80 | (PyTorch)</option>
81 |         </select>
82 |     </div>
83 |
84 |     { /* Forecast Horizon */ }
85 |     <div>
86 |         <label className="block text-xs font-semibold
87 | text-slate-700 mb-1.5 flex items-center gap-1.5">
88 |             <Calendar className="w-3.5 h-3.5 text-indigo-600" />
89 |             <span>Horizon Proyeksi (h Hari):</span>
90 |         </label>
91 |         <select
92 |             value={h}
93 |             onChange={(e) => setH(parseInt(e.target.value))}
94 |             className="w-full bg-slate-50 border border-slate-300
95 | rounded-lg px-3 py-2 text-xs font-mono text-slate-800
96 | focus:outline-none focus:border-emerald-600 cursor-pointer
97 | font-medium"
98 |         >
99 |             <option value={4}>4 Hari Ke Depan (Default
100 | Jurnal)</option>
101 |             <option value={7}>7 Hari Ke Depan (1 Minggu)</option>
102 |             <option value={14}>14 Hari Ke Depan (2 Minggu)</option>
103 |         </select>
104 |     </div>
105 |
106 |     { /* Execute Button */ }
107 |     <div>
108 |         <button
109 |             onClick={onRunPipeline}
110 |             disabled={isRunning}
111 |             className="w-full flex items-center justify-center gap-2
112 | bg-emerald-600 hover:bg-emerald-700 text-white font-bold px-4 py-2
113 | text-xs rounded-lg transition-colors disabled:opacity-50
114 | cursor-pointer shadow-sm"
115 |         >
116 |             <Play className={`w-3.5 h-3.5 ${isRunning ?
117 | 'animate-spin' : ''}`} />
118 |             <span>{isRunning ? 'Menjalankan Exec Engine...' :
119 | 'Jalankan Pipeline Auto'}</span>
120 |         </button>
121 |     </div>
122 | </div>
123 | );
124 | };

```

backend/main.py:

```

1 | import os
2 | import json
3 | import urllib.request
4 | import numpy as np
5 | import pandas as pd
6 | from typing import List, Optional
7 | from fastapi import FastAPI, HTTPException, Query
8 | from fastapi.middleware.cors import CORSMiddleware
9 | from pydantic import BaseModel, Field
10 |
11 | from peatfr_engine.pfvi import simulate_pfvi,
12 | optimize_pfvi_parameters, calculate_di_obs
13 | from peatfr_engine.imputation import impute_peatfr_data

```

```

13 | from peatfr_engine.forecasting import forecast_peatfr_variables
14 | from peatfr_engine.fire_intelligence import fire_engine
15 |
16 | app = FastAPI(
17 |     title="PeatFR EWS Engine API",
18 |     description="Early Warning System & Fire Vulnerability Index
    (PFVI) Forecasting API for Tropical Peatlands",
19 |     version="1.0.0"
20 | )
21 |
22 | app.add_middleware(
23 |     CORSMiddleware,
24 |     allow_origins=["*"],
25 |     allow_credentials=True,
26 |     allow_methods=["*"],
27 |     allow_headers=["*"],
28 | )
29 |
30 | # Load Indonesia Administrative Regions
31 | ADMIN_FILE = os.path.join(os.path.dirname(__file__),
    "peatfr_engine", "indonesia_admin.json")
32 | try:
33 |     with open(ADMIN_FILE) as f:
34 |         INDONESIA_REGIONS = json.load(f)
35 | except Exception:
36 |     INDONESIA_REGIONS = []
37 |
38 | class ImputeRequest(BaseModel):
39 |     WT: List[Optional[float]]
40 |     SM: List[Optional[float]]
41 |     Rf: List[Optional[float]]
42 |     Temp: List[Optional[float]]
43 |     method: str = Field("knn", description="imputation method: knn,
    spline, loess, linear")
44 |     k: int = Field(5, description="k parameter for knn")
45 |
46 | class ForecastRequest(BaseModel):
47 |     WT: List[float]
48 |     SM: List[float]
49 |     Rf: List[float]
50 |     Temp: List[float]
51 |     h: int = Field(4, description="forecasting horizon (days)")
52 |     model: str = Field("arima", description="model type: arima,
    lstm, gru")
53 |     look_back: int = Field(12, description="sliding window for
    LSTM/GRU")
54 |     hidden_units: int = Field(32, description="neurons count")
55 |     epochs: int = Field(50, description="training epochs")
56 |
57 | class AutoPipelineRequest(BaseModel):
58 |     WT: List[Optional[float]]
59 |     SM: List[Optional[float]]
60 |     Rf: List[Optional[float]]
61 |     Temp: List[Optional[float]]
62 |     imputation: str = Field("knn")
63 |     model: str = Field("arima")
64 |     h: int = Field(4)
65 |     r0: float = Field(3000.0)
66 |     look_back: int = Field(12)
67 |     hidden_units: int = Field(32)
68 |     epochs: int = Field(50)
69 |
70 | @app.get("/")
71 | @app.head("/")
72 | def read_root():
73 |     return {
74 |         "status": "online",
75 |         "service": "PeatFR Early Warning System Engine",
76 |         "version": "1.0.0",
77 |         "docs": "/docs"
78 |     }
79 |
80 | @app.get("/api/v1/health")
81 | @app.head("/api/v1/health")
82 | def health_check():
83 |     return {"status": "ok", "engine": "Python PyTorch & SciPy PeatFR
    Core"}
84 |
85 | @app.get("/api/v1/indonesia/regions")
86 | def get_indonesia_regions():
87 |     """Returns administrative hierarchy of Indonesia Peatland
    Provinces & Regencies."""
88 |     return {"status": "success", "provinces": INDONESIA_REGIONS}
89 |

```

```

90 | @app.get("/api/v1/realtime-peatland-data")
91 | def get_realtime_openmeteo_data(
92 |     lat: float = Query(-2.321, description="Latitude coordinate"),
93 |     lon: float = Query(113.901, description="Longitude coordinate"),
94 |     days: int = Query(30, description="Past days of historical open
    |     satellite data")
95 | ):
96 |     try:
97 |         url = (
98 |             f"https://api.open-meteo.com/v1/forecast?"
99 |             f"latitude={lat}&longitude={lon}&"
100 |             f"daily=temperature_2m_max,precipitation_sum&"
101 |             f"hourly=soil_moisture_0_to_7cm,soil_moisture_7_to_28cm&"
102 |             f"timezone=Asia%2FJakarta&past_days={days}&forecast_days=1"
103 |         )
104 |         req = urllib.request.Request(url, headers={"User-Agent":
    |         "Mozilla/5.0 (PeatFR-EWS/1.0)"})
105 |         res = urllib.request.urlopen(req, timeout=10)
106 |         data = json.loads(res.read().decode())
107 |
108 |         daily = data["daily"]
109 |         hourly = data["hourly"]
110 |
111 |         dates = daily["time"]
112 |         temp_max = daily["temperature_2m_max"]
113 |         precip = daily["precipitation_sum"]
114 |
115 |         sm_0_7 = np.array(hourly["soil_moisture_0_to_7cm"])
116 |         sm_7_28 = np.array(hourly["soil_moisture_7_to_28cm"])
117 |         sm_combined = (sm_0_7 * 0.4 + sm_7_28 * 0.6)
118 |
119 |         n_days = len(dates)
120 |         sm_raw = sm_combined[:n_days * 24].reshape(n_days,
    |         24).mean(axis=1)
121 |
122 |         sm_pct = np.clip(32.0 + ((sm_raw - 0.08) / (0.42 - 0.08)) *
    |         43.0, 30.0, 75.0)
123 |         wt_est = -0.30 - 0.012 * (65.0 - sm_pct) - 0.004 *
    |         np.maximum(0, 5.0 - np.array(precip))
124 |         wt_est = np.clip(wt_est, -1.8, -0.1)
125 |
126 |         # Multi-Source Fire Intelligence
127 |         fire_nearby = fire_engine.fetch_fireping_nearby(lat, lon,
    |         radius_m=25000)
128 |         latest_temp = float(temp_max[-1]) if temp_max else 34.0
129 |         latest_rf = float(precip[-1]) if precip else 0.0
130 |         latest_sm = float(sm_pct[-1]) if len(sm_pct) > 0 else 40.0
131 |         fwi_intel = fire_engine.fetch_owm_fwi(lat, lon,
    |         temp=latest_temp, rf=latest_rf, sm=latest_sm)
132 |
133 |         return {
134 |             "source": "Open-Meteo / ERA5-Land Realtime Satellite
    |             Data",
135 |             "lat": lat,
136 |             "lon": lon,
137 |             "dates": dates,
138 |             "WT": np.round(wt_est, 3).tolist(),
139 |             "SM": np.round(sm_pct, 2).tolist(),
140 |             "Rf": np.round(precip, 2).tolist(),
141 |             "Temp": np.round(temp_max, 2).tolist(),
142 |             "fire_intelligence": {
143 |                 "nearby": fire_nearby,
144 |                 "fwi": fwi_intel
145 |             }
146 |         }
147 |     except Exception as e:
148 |         raise HTTPException(status_code=500, detail=f"Failed to
    |         fetch realtime open data: {str(e)}")
149 |
150 | @app.get("/api/v1/fire-intelligence/nearby")
151 | def get_nearby_fires(
152 |     lat: float = Query(-2.321, description="Latitude"),
153 |     lon: float = Query(113.901, description="Longitude"),
154 |     radius_m: int = Query(25000, description="Radius in meters")
155 | ):
156 |     """Fetch nearby satellite fire detection & GWIS 7-day burned
    |     area (FirePing Public API)."""
157 |     return fire_engine.fetch_fireping_nearby(lat, lon,
    |     radius_m=radius_m)
158 |
159 | @app.get("/api/v1/fire-intelligence/gfw-peatland-fires")
160 | def get_gfw_peatland_fires(
161 |     limit: int = Query(100, description="Maximum fire alerts")
162 | ):

```

```

163 |     """Fetch vector NASA VIIRS active fire alerts specifically on
164 |     Indonesian peatlands (is__peatland = true)."""
165 |     return fire_engine.fetch_gfw_peatland_fires(limit=limit)
166 | @app.get("/api/v1/fire-intelligence/firms-hotspots")
167 | def get_firms_hotspots(
168 |     bbox: str = Query("95,-11,141,6", description="Bounding box
169 |     [min_lon,min_lat,max_lon,max_lat]"),
170 |     source: str = Query("VIIRS_SNPP_NRT", description="Satellite
171 |     sensor: VIIRS_SNPP_NRT, MODIS_NRT, VIIRS_NOAA20_NRT,
172 |     VIIRS_NOAA21_NRT"),
173 |     day_range: int = Query(1, ge=1, le=10, description="Range of
174 |     days back (1-10)")
175 | ):
176 |     """Fetch live NASA FIRMS active fire hotspots via authorized
177 |     MAP_KEY."""
178 |     return fire_engine.fetch_firms_hotspots(bbox=bbox,
179 |     source=source, day_range=day_range)
180 | @app.get("/api/v1/fire-intelligence/severe-alerts")
181 | def get_severe_fire_alerts():
182 |     """Fetch active severe fire alerts (Kebakaran Parah) across
183 |     Indonesian Peatlands."""
184 |     return fire_engine.fetch_severe_fire_alerts()
185 | @app.get("/api/v1/fire-intelligence/fwi")
186 | def get_fwi_forecast(
187 |     lat: float = Query(-2.321, description="Latitude"),
188 |     lon: float = Query(113.901, description="Longitude")
189 | ):
190 |     """Fetch Fire Weather Index (FWI) forecast (OpenWeatherMap /
191 |     Canadian FWI Engine)."""
192 |     return fire_engine.fetch_owm_fwi(lat, lon)
193 | @app.get("/api/v1/sabangau-sample")
194 | def get_sabangau_sample():
195 |     np.random.seed(42)
196 |     days = 192
197 |     time_idx = np.arange(days)
198 |
199 |     wt = -0.4 - 0.65 * (1.0 / (1.0 + np.exp(-(time_idx - 100) /
200 |     20.0))) + np.random.normal(0, 0.02, days)
201 |     sm = 65.0 - 27.0 * (1.0 / (1.0 + np.exp(-(time_idx - 100) /
202 |     20.0))) + np.random.normal(0, 1.2, days)
203 |     rf = np.maximum(0.0, np.random.exponential(8.0, days) * (1.0 -
204 |     (time_idx / 220.0)))
205 |     temp = 31.5 + 4.0 * (1.0 / (1.0 + np.exp(-(time_idx - 100) /
206 |     25.0))) + np.random.normal(0, 0.4, days)
207 |
208 |     wt_raw = wt.copy()
209 |     sm_raw = sm.copy()
210 |     rf_raw = rf.copy()
211 |     temp_raw = temp.copy()
212 |
213 |     wt_raw[114:117] = np.nan
214 |     sm_raw[114:117] = np.nan
215 |     rf_raw[114:117] = np.nan
216 |     temp_raw[145] = np.nan
217 |
218 |     return {
219 |         "location": "Sabangau, Central Kalimantan, Indonesia",
220 |         "time_steps": days,
221 |         "WT": [None if np.isnan(x) else round(float(x), 3) for x in
222 | wt_raw],
223 |         "SM": [None if np.isnan(x) else round(float(x), 2) for x in
224 | sm_raw],
225 |         "Rf": [None if np.isnan(x) else round(float(x), 2) for x in
226 | rf_raw],
227 |         "Temp": [None if np.isnan(x) else round(float(x), 2) for x
228 | in temp_raw]
229 |     }
230 | @app.post("/api/v1/impute")
231 | def impute_data(req: ImputeRequest):
232 |     try:
233 |         wt = np.array(req.WT, dtype=float)
234 |         sm = np.array(req.SM, dtype=float)
235 |         rf = np.array(req.Rf, dtype=float)
236 |         temp = np.array(req.Temp, dtype=float)
237 |
238 |         res = impute_peatfr_data(wt, sm, rf, temp,
239 |         method=req.method, k=req.k)
240 |
241 |         return {

```

```

229 |         "method": req.method,
230 |         "WT": res["WT"].tolist(),
231 |         "SM": res["SM"].tolist(),
232 |         "Rf": res["Rf"].tolist(),
233 |         "Temp": res["Temp"].tolist()
234 |     }
235 | except Exception as e:
236 |     raise HTTPException(status_code=400, detail=str(e))
237 |
238 | @app.post("/api/v1/forecast")
239 | def forecast_data(req: ForecastRequest):
240 |     try:
241 |         wt = np.array(req.WT, dtype=float)
242 |         sm = np.array(req.SM, dtype=float)
243 |         rf = np.array(req.Rf, dtype=float)
244 |         temp = np.array(req.Temp, dtype=float)
245 |
246 |         res = forecast_peatfr_variables(
247 |             wt, sm, rf, temp,
248 |             h=req.h,
249 |             model=req.model,
250 |             look_back=req.look_back,
251 |             hidden_units=req.hidden_units,
252 |             epochs=req.epochs
253 |         )
254 |
255 |         return {
256 |             "model": req.model,
257 |             "h": req.h,
258 |             "WT_pred": res["WT_pred"].tolist(),
259 |             "SM_pred": res["SM_pred"].tolist(),
260 |             "Rf_pred": res["Rf_pred"].tolist(),
261 |             "Temp_pred": res["Temp_pred"].tolist()
262 |         }
263 |     except Exception as e:
264 |         raise HTTPException(status_code=400, detail=str(e))
265 |
266 | @app.post("/api/v1/pipeline/auto")
267 | def run_auto_pipeline(req: AutoPipelineRequest):
268 |     try:
269 |         wt_in = np.array(req.WT, dtype=float)
270 |         sm_in = np.array(req.SM, dtype=float)
271 |         rf_in = np.array(req.Rf, dtype=float)
272 |         temp_in = np.array(req.Temp, dtype=float)
273 |
274 |         has_nans = any(np.isnan(wt_in)) or any(np.isnan(sm_in)) or
any(np.isnan(rf_in)) or any(np.isnan(temp_in))
275 |
276 |         if has_nans:
277 |             imp_res = impute_peatfr_data(wt_in, sm_in, rf_in,
temp_in, method=req.imputation)
278 |             wt_clean = imp_res["WT"]
279 |             sm_clean = imp_res["SM"]
280 |             rf_clean = imp_res["Rf"]
281 |             temp_clean = imp_res["Temp"]
282 |         else:
283 |             wt_clean, sm_clean, rf_clean, temp_clean = wt_in, sm_in,
rf_in, temp_in
284 |
285 |         fc_res = forecast_peatfr_variables(
286 |             wt_clean, sm_clean, rf_clean, temp_clean,
287 |             h=req.h,
288 |             model=req.model,
289 |             look_back=req.look_back,
290 |             hidden_units=req.hidden_units,
291 |             epochs=req.epochs
292 |         )
293 |
294 |         wt_full = np.concatenate([wt_clean, fc_res["WT_pred"]])
295 |         sm_full = np.concatenate([sm_clean, fc_res["SM_pred"]])
296 |         rf_full = np.concatenate([rf_clean, fc_res["Rf_pred"]])
297 |         temp_full = np.concatenate([temp_clean,
fc_res["Temp_pred"]])
298 |
299 |         opt_params, opt_mse = optimize_pfvi_parameters(wt_clean,
sm_clean, rf_clean, temp_clean, r0=req.r0)
300 |         a_h, b_h, n, alpha = opt_params
301 |
302 |         pfvi_sim = simulate_pfvi(wt_full, sm_full, rf_full,
temp_full, a_h, b_h, n, alpha, r0=req.r0)
303 |
304 |         pfvi_series = pfvi_sim["pfvi"].tolist()
305 |         pfvi_forecast = pfvi_series[-req.h:]
306 |

```

```

307 |         risk_categories = []
308 |         for val in pfvi_forecast:
309 |             if val < 75:
310 |                 risk_categories.append("Low")
311 |             elif val < 150:
312 |                 risk_categories.append("Moderate")
313 |             elif val < 225:
314 |                 risk_categories.append("High")
315 |             else:
316 |                 risk_categories.append("Extreme")
317 |
318 |         current_status = risk_categories[-1] if risk_categories else
319 |         "Unknown"
320 |
321 |         return {
322 |             "status": "success",
323 |             "has_missing_data_imputed": has_nans,
324 |             "imputation_method": req.imputation,
325 |             "forecasting_model": req.model,
326 |             "optimization": {
327 |                 "a_h": float(a_h),
328 |                 "b_h": float(b_h),
329 |                 "n": float(n),
330 |                 "alpha": float(alpha),
331 |                 "mse": float(opt_mse)
332 |             },
333 |             "historical_len": len(wt_clean),
334 |             "forecast_horizon": req.h,
335 |             "forecast": {
336 |                 "WT": fc_res["WT_pred"].tolist(),
337 |                 "SM": fc_res["SM_pred"].tolist(),
338 |                 "Rf": fc_res["Rf_pred"].tolist(),
339 |                 "Temp": fc_res["Temp_pred"].tolist(),
340 |                 "PFVI": pfvi_forecast,
341 |                 "Risk_Categories": risk_categories,
342 |                 "Current_Status": current_status
343 |             },
344 |             "full_series": {
345 |                 "WT": wt_full.tolist(),
346 |                 "SM": sm_full.tolist(),
347 |                 "Rf": rf_full.tolist(),
348 |                 "Temp": temp_full.tolist(),
349 |                 "PFVI": pfvi_series,
350 |                 "DI_obs": pfvi_sim["di_obs"].tolist(),
351 |                 "DF": pfvi_sim["df"].tolist(),
352 |                 "RF": pfvi_sim["rf"].tolist(),
353 |                 "WTF": pfvi_sim["wtf"].tolist()
354 |             }
355 |         }
356 |     except Exception as e:
357 |         raise HTTPException(status_code=400, detail=str(e))

```

Screenshot Aplikasi:

[Screenshot: Pipeline Controls: imputasi, model AI, dan horizon proyeksi — ss_pipeline.png]

FITUR 5: Grafik Historis & Proyeksi Time-Series (Forecast Charts)

Deskripsi & Tujuan Fitur: Visualisasi recharts interaktif: grafik utama **PFVI observasi vs forecast** (garis hijau solid vs garis merah putus-putus) dengan zona ambang berwarna (Extreme ≥ 225 , High 150–225, Moderate 75–150), garis referensi threshold, dan grafik pendukung deret **WT, SM, Rf, Temp** (historis + proyeksi). Data berasal dari `full_series` hasil pipeline (DF, RF, WTF, DI_obs, PFVI).

Cara Menggunakan:

1. Amati grafik PFVI: bagian kiri (hijau) = observasi historis, bagian kanan (merah putus-putus) = proyeksi ke depan.
1. Hover pada grafik untuk melihat tooltip nilai per hari (step H-.../H+...).
1. Cek grafik pendukung WT/SM/Rf/Temp untuk melihat lintasan proyeksi tiap variabel hidrologi.

Kode Sumber (Source Code Listing): frontend/src/components/ForecastCharts.tsx

```

1 | import React from 'react';
2 | import {
3 |   ResponsiveContainer,
4 |   ComposedChart,
5 |   LineChart,
6 |   Line,
7 |   XAxis,
8 |   YAxis,
9 |   CartesianGrid,
10 |   Tooltip,
11 |   ReferenceLine,
12 |   ReferenceArea
13 | } from 'recharts';
14 | import { Activity } from 'lucide-react';
15 |
16 | interface ForecastChartsProps {
17 |   fullSeries: {
18 |     WT: number[];
19 |     SM: number[];
20 |     Rf: number[];
21 |     Temp: number[];
22 |     PFVI: number[];
23 |     DI_obs: number[];
24 |   };
25 |   forecastHorizon: number;
26 | }
27 |
28 | export const ForecastCharts: React.FC<ForecastChartsProps> = ({
29 |   fullSeries, forecastHorizon }) => {
30 |   const totalPoints = fullSeries.PFVI.length;
31 |   const historicalCount = totalPoints - forecastHorizon;
32 |
33 |   // Prepare chart dataset
34 |   const chartData = fullSeries.PFVI.map((pfviVal, idx) => {
35 |     const isForecast = idx >= historicalCount;
36 |     const dayLabel = isForecast ? `H+${idx - historicalCount + 1}` :
37 |       `H-${historicalCount - idx}`;
38 |
39 |     return {
40 |       day: dayLabel,
41 |       step: idx + 1,
42 |       isForecast,
43 |       PFVI_hist: !isForecast || idx === historicalCount - 1 ?
44 |         pfviVal : null,
45 |       PFVI_pred: isForecast ? pfviVal : null,
46 |       DI_obs: fullSeries.DI_obs ? fullSeries.DI_obs[idx] : null,
47 |       WT: fullSeries.WT[idx],
48 |       SM: fullSeries.SM[idx],
49 |       Rf: fullSeries.Rf[idx],
50 |       Temp: fullSeries.Temp[idx],
51 |     };
52 |   });
53 |
54 |   return (
55 |     <div className="space-y-6 mb-6">
56 |       <div className="telemetry-panel bg-white border
57 |         border-slate-200 rounded-xl p-6 shadow-sm">
58 |         <div className="flex flex-col sm:flex-row items-start
59 |           sm:items-center justify-between gap-2 mb-6">
60 |           <div>
61 |             <div className="flex items-center gap-2">
62 |               <Activity className="w-5 h-5 text-emerald-600" />
63 |               <h3 className="text-base font-bold text-slate-900
64 |                 tracking-wide">
65 |                 Grafik Historis & Forecast Peat Fire Vulnerability
66 |                 Index (PFVI)
67 |               </h3>
68 |             </div>
69 |             <p className="text-xs text-slate-500 font-medium mt-1">
70 |               Garis hijau solid = Data Observasi/Historis | Garis
71 |               merah putus-putus = Proyeksi {forecastHorizon} Hari Ke Depan
72 |             </p>
73 |           </div>
74 |           <div className="flex items-center gap-3 text-xs font-mono
75 |             font-semibold">
76 |             <span className="flex items-center gap-1
77 |               text-emerald-700">
78 |               <span className="w-3 h-1 bg-emerald-600 inline-block
79 |                 rounded" /> PFVI Observasi
80 |             </span>
81 |             <span className="flex items-center gap-1 text-red-600">
82 |               <span className="w-3 h-1 bg-red-600 border-t

```

```

border-dashed border-red-600 inline-block rounded" /> Forecast
74 |     </span>
75 |   </div>
76 | </div>
77 |
78 |   <div className="h-[340px] w-full">
79 |     <ResponsiveContainer width="100%" height="100%">
80 |       <ComposedChart data={chartData} margin={{ top: 10,
right: 20, left: -10, bottom: 0 }}>
81 |         <CartesianGrid strokeDasharray="3 3" stroke="#e2e8f0"
vertical={false} />
82 |         <XAxis dataKey="day" stroke="#64748b" tick={{
fontSize: 11 }} />
83 |         <YAxis domain={[0, 300]} stroke="#64748b" tick={{
fontSize: 11 }} />
84 |         <Tooltip
85 |           contentStyle={{
86 |             backgroundColor: 'ffffff',
87 |             borderColor: '#cbd5e1',
88 |             borderRadius: '8px',
89 |             fontSize: '12px',
90 |             color: '#0f172a',
91 |             boxShadow: '0 4px 6px -1px rgba(0, 0, 0, 0.1)'
92 |           }}
93 |         />
94 |
95 |         {/* Shaded Threshold Zones */}
96 |         <ReferenceArea y1={225} y2={300} fill="#dc2626"
fillOpacity={0.08} />
97 |         <ReferenceArea y1={150} y2={225} fill="#ea580c"
fillOpacity={0.08} />
98 |         <ReferenceArea y1={75} y2={150} fill="#d97706"
fillOpacity={0.05} />
99 |
100 |         <ReferenceLine y={225} stroke="#dc2626"
strokeDasharray="3 3" label={{ value: 'Extreme (225)', fill:
'#dc2626', fontSize: 10, fontWeight: 'bold' }} />
101 |         <ReferenceLine y={150} stroke="#ea580c"
strokeDasharray="3 3" label={{ value: 'High (150)', fill: '#ea580c',
fontSize: 10, fontWeight: 'bold' }} />
102 |         <ReferenceLine y={75} stroke="#d97706"
strokeDasharray="3 3" label={{ value: 'Moderate (75)', fill:
'#d97706', fontSize: 10, fontWeight: 'bold' }} />
103 |
104 |         <Line type="monotone" dataKey="PFVI_hist"
stroke="#16a34a" strokeWidth={2.5} dot={false} name="PFVI Historis"
/>
105 |         <Line type="monotone" dataKey="PFVI_pred"
stroke="#dc2626" strokeWidth={2.5} strokeDasharray="5 5" dot={{ r:
4, fill: '#dc2626' }} name="PFVI Forecast" />
106 |       </ComposedChart>
107 |     </ResponsiveContainer>
108 |   </div>
109 | </div>
110 |
111 |   {/* 4 Parameter Synchronized Grid Charts */}
112 |   <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
113 |     {/* Water Table Chart */}
114 |     <div className="telemetry-panel bg-white border
border-slate-200 rounded-xl p-5 shadow-sm">
115 |       <h4 className="text-xs font-bold text-slate-800 uppercase
tracking-wider mb-3 flex items-center justify-between">
116 |         <span>Kedalaman Muka Air Tanah (WT - meter)</span>
117 |         <span className="text-cyan-700 font-mono">WT</span>
118 |       </h4>
119 |       <div className="h-[180px] w-full">
120 |         <ResponsiveContainer width="100%" height="100%">
121 |           <LineChart data={chartData}>
122 |             <CartesianGrid strokeDasharray="3 3"
stroke="#e2e8f0" vertical={false} />
123 |             <XAxis dataKey="day" stroke="#64748b" tick={{
fontSize: 10 }} />
124 |             <YAxis stroke="#64748b" tick={{ fontSize: 10 }}
domain={['auto', 'auto']} />
125 |             <Tooltip contentStyle={{ backgroundColor: 'ffffff',
borderColor: '#cbd5e1', borderRadius: '8px', fontSize: '11px',
color: '#0f172a' }} />
126 |             <Line type="monotone" dataKey="WT" stroke="#0891b2"
strokeWidth={2} dot={false} />
127 |           </LineChart>
128 |         </ResponsiveContainer>
129 |       </div>
130 |     </div>
131 |

```

```

132 |         { /* Soil Moisture Chart */
133 |         <div className="telemetry-panel bg-white border
border-slate-200 rounded-xl p-5 shadow-sm">
134 |             <h4 className="text-xs font-bold text-slate-800 uppercase
tracking-wider mb-3 flex items-center justify-between">
135 |                 <span>Kelembaban Tanah (SM - %)</span>
136 |                 <span className="text-blue-700 font-mono">SM</span>
137 |             </h4>
138 |             <div className="h-[180px] w-full">
139 |                 <ResponsiveContainer width="100%" height="100%">
140 |                     <LineChart data={chartData}>
141 |                         <CartesianGrid strokeDasharray="3 3"
stroke="#e2e8f0" vertical={false} />
142 |                         <XAxis dataKey="day" stroke="#64748b" tick={{
fontSize: 10 }} />
143 |                         <YAxis stroke="#64748b" tick={{ fontSize: 10 }}
domain=[[30, 80]] />
144 |                         <Tooltip contentStyle={{ backgroundColor: 'ffffff',
borderColor: '#cbd5e1', borderRadius: '8px', fontSize: '11px',
color: '#0f172a' }} />
145 |                         <Line type="monotone" dataKey="SM" stroke="#2563eb"
strokeWidth={2} dot={false} />
146 |                     </LineChart>
147 |                 </ResponsiveContainer>
148 |             </div>
149 |         </div>
150 |
151 |         { /* Rainfall Chart */
152 |         <div className="telemetry-panel bg-white border
border-slate-200 rounded-xl p-5 shadow-sm">
153 |             <h4 className="text-xs font-bold text-slate-800 uppercase
tracking-wider mb-3 flex items-center justify-between">
154 |                 <span>Curah Hujan (Rf - mm/hari)</span>
155 |                 <span className="text-indigo-700 font-mono">Rf</span>
156 |             </h4>
157 |             <div className="h-[180px] w-full">
158 |                 <ResponsiveContainer width="100%" height="100%">
159 |                     <LineChart data={chartData}>
160 |                         <CartesianGrid strokeDasharray="3 3"
stroke="#e2e8f0" vertical={false} />
161 |                         <XAxis dataKey="day" stroke="#64748b" tick={{
fontSize: 10 }} />
162 |                         <YAxis stroke="#64748b" tick={{ fontSize: 10 }}
domain=[[0, 'auto']] />
163 |                         <Tooltip contentStyle={{ backgroundColor: 'ffffff',
borderColor: '#cbd5e1', borderRadius: '8px', fontSize: '11px',
color: '#0f172a' }} />
164 |                         <Line type="monotone" dataKey="Rf" stroke="#4f46e5"
strokeWidth={2} dot={false} />
165 |                     </LineChart>
166 |                 </ResponsiveContainer>
167 |             </div>
168 |         </div>
169 |
170 |         { /* Air Temp Chart */
171 |         <div className="telemetry-panel bg-white border
border-slate-200 rounded-xl p-5 shadow-sm">
172 |             <h4 className="text-xs font-bold text-slate-800 uppercase
tracking-wider mb-3 flex items-center justify-between">
173 |                 <span>Suhu Udara Maksimum (Temp - °C)</span>
174 |                 <span className="text-rose-700 font-mono">Temp</span>
175 |             </h4>
176 |             <div className="h-[180px] w-full">
177 |                 <ResponsiveContainer width="100%" height="100%">
178 |                     <LineChart data={chartData}>
179 |                         <CartesianGrid strokeDasharray="3 3"
stroke="#e2e8f0" vertical={false} />
180 |                         <XAxis dataKey="day" stroke="#64748b" tick={{
fontSize: 10 }} />
181 |                         <YAxis stroke="#64748b" tick={{ fontSize: 10 }}
domain=[[28, 40]] />
182 |                         <Tooltip contentStyle={{ backgroundColor: 'ffffff',
borderColor: '#cbd5e1', borderRadius: '8px', fontSize: '11px',
color: '#0f172a' }} />
183 |                         <Line type="monotone" dataKey="Temp"
stroke="#e11d48" strokeWidth={2} dot={false} />
184 |                     </LineChart>
185 |                 </ResponsiveContainer>
186 |             </div>
187 |         </div>
188 |     </div>
189 | </div>
190 | );
191 | };

```

Screenshot Aplikasi:*[Screenshot: Grafik historis & proyeksi PFVI beserta deret WT/SM/Rf/Temp — ss_charts.png]***FITUR 6: Simulator Skenario Cuaca What-If (Scenario Simulator)**

Deskripsi & Tujuan Fitur: Alat simulasi hipotetis di sisi klien: pengguna menggeser empat slider — **Kenaikan Suhu** (+0...+5 °C), **Durasi Kemarau** (1–30 hari tanpa hujan), **Kedalaman Muka Air Tanah** (0.2–1.8 m), dan **PFVI Baseline Awal** (20–250) — lalu aplikasi menghitung ulang PFVI akhir setelah N hari kemarau menggunakan persamaan fisik PeatFR (DF evapotranspirasi, RF = 0, WTF van Genuchten dengan parameter kalibrasi Sabangau: $a_H=6.5$, $b_H=0.02$, $n=18.2$, $\alpha=0.9$). Hasil menampilkan **PFVI Prediksi**, laju DF, dan WTF.

Cara Menggunakan:

1. Geser slider **Kenaikan Suhu Udara** untuk mensimulasikan gelombang panas.
1. Geser slider **Durasi Kemarau Tanpa Hujan** untuk jangka waktu kekeringan.
1. Geser slider **Kedalaman Muka Air Tanah** — semakin dalam TMA, semakin besar kapilaritas yang hilang (WTF).
1. Atur **PFVI Baseline Awal** lalu baca kotak hasil: PFVI prediksi setelah N hari dan komponen DF/WTF per hari.

Kode Sumber (Source Code Listing): frontend/src/components/ScenarioSimulator.tsx

```

1 | import React, { useState } from 'react';
2 | import { Sliders, Flame, AlertTriangle } from 'lucide-react';
3 |
4 | export const ScenarioSimulator: React.FC = () => {
5 |   const [tempDelta, setTempDelta] = useState<number>(2.0);
6 |   const [dryDays, setDryDays] = useState<number>(10);
7 |   const [wtDepth, setWtDepth] = useState<number>(1.1); // meters
8 |   const [initialPfvi, setInitialPfvi] = useState<number>(180.0);
9 |
10 |   // Simulates hypothetical scenario
11 |   const simulateScenario = () => {
12 |     let current = initialPfvi;
13 |     const baseTemp = 33.5 + tempDelta;
14 |     const wtCm = wtDepth * 100.0;
15 |
16 |     // Calibration parameters (typical Sabangau calibrated values)
17 |     const aH = 6.5;
18 |     const bH = 0.02;
19 |     const n = 18.2;
20 |     const alpha = 0.9;
21 |
22 |     const timeline = [];
23 |     for (let day = 1; day <= dryDays; day++) {
24 |       const df = (300.0 - current) * (0.4982 * Math.exp(0.0905 *
25 |         baseTemp + 1.6096) - 4.268) * 1e-3 / (1.0 + 10.88 *
26 |         Math.exp(-0.0017358 * 3000));
27 |       const rf = 0.0; // 0 mm rain during dry spell
28 |
29 |       const m = 1.0 - 1.0 / n;
30 |       const theta = Math.pow(1.0 + Math.pow(wtCm / alpha, n), -m);
31 |       const wtf = aH - bH * (1.0 - theta) * 300.0;
32 |
33 |       current = Math.min(300.0, Math.max(0.0, current + df - rf -
34 |         wtf));
35 |       timeline.push({ day, pfvi: current, df, wtf });
36 |     }
37 |     return { finalPfvi: current, timeline };
38 |   };
39 |
40 |   const simResult = simulateScenario();
41 |
42 |   return (
    <div className="telemetry-panel bg-white border border-slate-200
    rounded-xl p-6 mb-6 shadow-sm">
      <div className="flex items-center justify-between mb-6
      border-b border-slate-200 pb-4">
        <div className="flex items-center gap-2">

```

```

43 |         <Sliders className="w-5 h-5 text-amber-600" />
44 |         <div>
45 |             <h3 className="text-base font-bold text-slate-900
tracking-wide">
46 |                 Simulasi Skenario Cuaca & Respon Kerawanan Lahan
("What-If Scenario")
47 |             </h3>
48 |             <p className="text-xs text-slate-500 font-medium">
49 |                 Uji dampak kenaikan suhu, periode kemarau, dan
penurunan muka air tanah terhadap indeks PFVI
50 |             </p>
51 |         </div>
52 |     </div>
53 | </div>
54 |
55 |     <div className="grid grid-cols-1 lg:grid-cols-12 gap-6">
56 |         {/ * Sliders Input Controls */}
57 |         <div className="lg:col-span-7 space-y-5">
58 |             {/ * Temp Delta Slider */}
59 |             <div>
60 |                 <div className="flex justify-between text-xs font-mono
font-semibold text-slate-700 mb-1.5">
61 |                     <span>Kenaikan Suhu Udara (Temp Delta):</span>
62 |                     <span className="text-rose-600
font-bold">+{tempDelta.toFixed(1)} °C</span>
63 |                 </div>
64 |                 <input
65 |                     type="range"
66 |                     min="0"
67 |                     max="5"
68 |                     step="0.5"
69 |                     value={tempDelta}
70 |                     onChange={(e) =>
setTempDelta(parseFloat(e.target.value))}
71 |                     className="w-full accent-rose-600 cursor-pointer"
72 |                 />
73 |             </div>
74 |
75 |             {/ * Dry Spell Slider */}
76 |             <div>
77 |                 <div className="flex justify-between text-xs font-mono
font-semibold text-slate-700 mb-1.5">
78 |                     <span>Durasi Kemarau Tanpa Hujan (Hari):</span>
79 |                     <span className="text-amber-600 font-bold">{dryDays}
Hari</span>
80 |                 </div>
81 |                 <input
82 |                     type="range"
83 |                     min="1"
84 |                     max="30"
85 |                     step="1"
86 |                     value={dryDays}
87 |                     onChange={(e) => setDryDays(parseInt(e.target.value))}
88 |                     className="w-full accent-amber-600 cursor-pointer"
89 |                 />
90 |             </div>
91 |
92 |             {/ * Water Table Depth Slider */}
93 |             <div>
94 |                 <div className="flex justify-between text-xs font-mono
font-semibold text-slate-700 mb-1.5">
95 |                     <span>Kedalaman Muka Air Tanah (WT Depth Below
Surface):</span>
96 |                     <span className="text-cyan-700
font-bold">-{wtDepth.toFixed(2)} m</span>
97 |                 </div>
98 |                 <input
99 |                     type="range"
100 |                     min="0.2"
101 |                     max="1.8"
102 |                     step="0.05"
103 |                     value={wtDepth}
104 |                     onChange={(e) =>
setWtDepth(parseFloat(e.target.value))}
105 |                     className="w-full accent-cyan-600 cursor-pointer"
106 |                 />
107 |             </div>
108 |
109 |             {/ * Initial PFVI */}
110 |             <div>
111 |                 <div className="flex justify-between text-xs font-mono
font-semibold text-slate-700 mb-1.5">
112 |                     <span>PFVI Baseline Awal:</span>
113 |                     <span className="text-slate-900

```

```

font-bold">{initialPfvi.toFixed(1)}</span>
114 | </div>
115 | <input
116 |   type="range"
117 |   min="20"
118 |   max="250"
119 |   step="5"
120 |   value={initialPfvi}
121 |   onChange={(e) =>
setInitialPfvi(parseFloat(e.target.value))}
122 |   className="w-full accent-slate-600 cursor-pointer"
123 |   />
124 | </div>
125 | </div>
126 |
127 |   { /* Simulation Result Readout Box */
128 |   <div className="lg:col-span-5 bg-slate-50 p-5 rounded-xl
border border-slate-200 flex flex-col justify-between">
129 |     <div>
130 |       <div className="text-xs font-mono font-bold uppercase
tracking-wider text-slate-500 mb-2 flex items-center
justify-between">
131 |         <span>Hasil Proyeksi Skenario</span>
132 |         <AlertTriangle className="w-4 h-4 text-amber-600" />
133 |       </div>
134 |
135 |       <div className="space-y-4 my-3">
136 |         <div>
137 |           <div className="text-xs text-slate-500
font-medium">PFVI Awal:</div>
138 |           <div className="text-xl font-mono font-bold
text-slate-800">{initialPfvi.toFixed(1)} / 300.0</div>
139 |         </div>
140 |
141 |         <div>
142 |           <div className="text-xs text-slate-500
font-medium">PFVI Prediksi Setelah {dryDays} Hari:</div>
143 |           <div className="text-3xl font-mono font-extrabold
text-red-600">
144 |             {simResult.finalPfvi.toFixed(1)} <span
className="text-xs text-slate-500 font-normal">/ 300.0</span>
145 |           </div>
146 |         </div>
147 |       </div>
148 |     </div>
149 |
150 |     <div className="pt-3 border-t border-slate-200 text-[11px]
text-slate-600 font-medium space-y-1">
151 |       <div className="flex justify-between">
152 |         <span>Laju Evapotranspirasi (DF):</span>
153 |         <span className="font-mono font-bold
text-slate-800">+{simResult.timeline[0]?.df.toFixed(2)} /hr</span>
154 |       </div>
155 |       <div className="flex justify-between">
156 |         <span>Kapilaritas Air Tanah (WTF):</span>
157 |         <span className="font-mono font-bold
text-slate-800">-{simResult.timeline[0]?.wtf.toFixed(2)} /hr</span>
158 |       </div>
159 |     </div>
160 |   </div>
161 | </div>
162 | </div>
163 |   );
164 | };

```

Screenshot Aplikasi:

[Screenshot: What-If Simulator: pengujian skenario suhu, kemarau, dan kedalaman TMA — ss_simulator.png]

FITUR 7: Modal Spesifikasi Teoretis (Metadata Jurnal & Integrasi API Satelit)

Deskripsi & Tujuan Fitur: Modal ringkasan metodologi: informasi publikasi jurnal PeatFR (*Ecological Informatics* 92 (2025) 103532, Elsevier, DOI 10.1016/j.ecoinf.2025.103532; penulis: Mahdiyasa, Melly, Pasaribu (ITB), Taufik (IPB), Muljadi (Univ. Nottingham)) dan daftar integrasi open satellite & hotspot data sources (NASA FIRMS, GIBS, GFW, Open-Meteo).

Cara Menggunakan:

1. Klik tombol **Spesifikasi Teoretis** di header.
1. Baca metadata jurnal (judul, penulis, jurnal, DOI) pada panel pertama.
1. Gulir daftar integrasi API satelit pada panel kedua.
1. Tekan tombol **X** atau area gelap di luar modal untuk menutup.

Kode Sumber (Source Code Listing): frontend/src/components/TheoreticalSpecsModal.tsx

```

1 | import React from 'react';
2 | import { X, BookOpen, Satellite, Flame } from 'lucide-react';
3 |
4 | interface TheoreticalSpecsModalProps {
5 |   isOpen: boolean;
6 |   onClose: () => void;
7 | }
8 |
9 | export const TheoreticalSpecsModal:
10 |   React.FC<TheoreticalSpecsModalProps> = ({ isOpen, onClose }) => {
11 |     if (!isOpen) return null;
12 |
13 |     return (
14 |       <div className="fixed inset-0 z-[100] flex items-center
15 |         justify-center p-4 bg-black/80 backdrop-blur-sm">
16 |         <div className="glass-panel w-full max-w-3xl max-h-[85vh]
17 |           overflow-y-auto rounded-2xl p-6 border border-white/10
18 |           text-slate-200">
19 |           <div className="flex items-center justify-between border-b
20 |             border-white/10 pb-4 mb-4">
21 |             <div className="flex items-center gap-2">
22 |               <BookOpen className="w-5 h-5 text-emerald-400" />
23 |               <h3 className="text-base font-bold text-white
24 |                 tracking-wide">
25 |                 Metodologi Jurnal & Integrasi Open Satellite APIs
26 |               </h3>
27 |             </div>
28 |             <button
29 |               onClick={onClose}
30 |               className="p-1 rounded-lg hover:bg-slate-800
31 |               text-slate-400 hover:text-white transition-colors"
32 |             >
33 |               <X className="w-5 h-5" />
34 |             </button>
35 |           </div>
36 |
37 |           <div className="space-y-4 text-xs leading-relaxed
38 |             text-slate-300">
39 |             <div className="bg-slate-900/60 p-4 rounded-xl border
40 |               border-white/5">
41 |               <h4 className="font-bold text-emerald-400 mb-1
42 |                 text-sm">Informasi Publikasi Jurnal</h4>
43 |               <p><strong>Judul:</strong> Peatfr: An R package to
44 |                 forecast tropical peatland fire risk with stochastic, machine
45 |                 learning, and optimisation methods</p>
46 |               <p><strong>Penulis:</strong> Adilan W. Mahdiyasa, Melly,
47 |                 Udjianna S. Pasaribu (ITB), Muh Taufik (IPB), Bagus P. Muljadi
48 |                 (Univ. of Nottingham)</p>
49 |               <p><strong>Jurnal:</strong> Ecological Informatics 92
50 |                 (2025) 103532, Elsevier</p>
51 |               <p><strong>DOI:</strong> <a
52 |                 href="https://doi.org/10.1016/j.ecoinf.2025.103532" target="_blank"
53 |                 rel="noreferrer" className="text-cyan-400
54 |                 underline">10.1016/j.ecoinf.2025.103532</a></p>
55 |             </div>
56 |
57 |             <div className="bg-slate-900/60 p-4 rounded-xl border
58 |               border-white/5 space-y-2">
59 |               <h4 className="font-bold text-amber-400 text-sm flex
60 |                 items-center gap-1.5">
61 |                 <Satellite className="w-4 h-4" /> Integrasi Open
62 |                 Satellite & Hotspot Data Sources
63 |               </h4>
64 |               <ul className="list-disc pl-5 space-y-1">
65 |                 <li>
66 |                   <strong>NASA FIRMS (Fire Information for Resource
67 |                     Management System):</strong> Overlay WMS real-time deteksi titik
68 |                     panas thermal VIIRS 375m (SNPP & NOAA-20) dalam 24 jam terakhir di
69 |                     seluruh wilayah Indonesia.

```

```

46 |         </li>
47 |         <li>
48 |             <strong>Open-Meteo & ECMWF ERA5-Land
Reanalysis:</strong> Data telemetri real-time harian untuk
temperatur maksimum, curah hujan, dan kelembaban tanah (0-7cm &
7-28cm) di seluruh stasiun gambut Indonesia.
49 |         </li>
50 |         <li>
51 |             <strong>Global Forest Watch (GFW) & OpenEPI
APIs:</strong> Dataset sejarah kebakaran dan peta tutupan lahan
gambut tropis Indonesia.
52 |         </li>
53 |     </ul>
54 | </div>
55 |
56 |     <div>
57 |         <h4 className="font-bold text-white text-sm mb-2">1.
Formula Peat Fire Vulnerability Index (PFVI)</h4>
58 |         <p className="mb-2">PFVI mengintegrasikan neraca air
atmosferik, kelembaban tanah, dan kapilaritas air tanah:</p>
59 |         <div className="font-mono bg-slate-950 p-3 rounded-lg
border border-white/10 text-emerald-300">
60 |             {"PFVI_t = PFVI_{t-1} + DF_t - RF_t - WTF_t"}
61 |         </div>
62 |         <ul className="list-disc pl-5 space-y-1 mt-2">
63 |             <li><strong>DF_t (Evapotranspiration Loss):</strong>
Fungsi dari temperatur maksimum udara (T_m) dan curah hujan tahunan
(R_0).</li>
64 |             <li><strong>RF_t (Rainfall Factor):</strong>
Mengakomodasi intersepsi kanopi awal (ambang 5.1 mm/hari).</li>
65 |             <li><strong>WTF_t (Water Table Factor):</strong>
Menggunakan fungsi retensi tanah van Genuchten  $\theta(v) = [1 +
(v/\alpha)^n]^{-m}$ .</li>
66 |         </ul>
67 |     </div>
68 |
69 |     <div>
70 |         <h4 className="font-bold text-white text-sm mb-2">2.
Optimisasi Parameter Nelder-Mead</h4>
71 |         <p>
72 |             Parameter  $a_H$ ,  $b_H$ ,  $\alpha$ ,  $n$  dikalibrasi secara dinamis
tanpa derivatif untuk meminimalkan Mean Squared Error (MSE) antara
PFVI simulasi dengan Indeks Kekeringan Observasi (DI_obs) yang
dihitung dari sensor kelembaban tanah.
73 |         </p>
74 |     </div>
75 | </div>
76 |
77 |     <div className="mt-6 pt-4 border-t border-white/10 flex
justify-end">
78 |         <button
79 |             onClick={onClose}
80 |             className="px-4 py-2 bg-slate-800 hover:bg-slate-700
text-white rounded-lg text-xs font-medium"
81 |         >
82 |             Tutup
83 |         </button>
84 |     </div>
85 | </div>
86 | </div>
87 | );
88 | };

```

Screenshot Aplikasi:

[Screenshot: Modal Spesifikasi Teoretis: metadata jurnal & integrasi API satelit — ss_modal.png]

FITUR 8: Backend API Routing & Telemetri Satelit Realtime (FastAPI Core)

Deskripsi & Tujuan Fitur: Server FastAPI (container `peatfr-api`) menyediakan 13 endpoint REST (lihat Bab 4.B) termasuk proxy telemetri Open-Meteo yang mengestimasi deret **WT** (dari soil moisture & presipitasi), **SM** (kombinasi bobot 0–7cm & 7–28cm), **Rf**, dan **Temp** dengan timezone Asia/Jakarta, lalu menggabungkan Fire Intelligence (FirePing nearby + FWI). Endpoint `/api/v1/pipeline/auto` adalah orkestrator pipeline lengkap: deteksi NaN → imputasi → forecasting → optimasi Nelder-Mead → simulasi PFVI → kategori risiko.

Cara Menggunakan (Developer/API Consumer):

1. Jalankan container atau akses `https://peatfr.bagaswibowo.app/api/v1/health` untuk cek status.
1. Panggil `GET /api/v1/realtime-peatland-data?lat=-2.321&lon;=113.901&days;=30` untuk deret telemetri + fire intelligence.
1. Kirim `POST /api/v1/pipeline/auto` dengan JSON {WT, SM, Rf, Temp, imputation, model, h, ...} untuk pipeline penuh.
1. Dokumentasi interaktif tersedia di `/docs` (Swagger UI).

Kode Sumber (Source Code Listing): backend/main.py

```

1 | import os
2 | import json
3 | import urllib.request
4 | import numpy as np
5 | import pandas as pd
6 | from typing import List, Optional
7 | from fastapi import FastAPI, HTTPException, Query
8 | from fastapi.middleware.cors import CORSMiddleware
9 | from pydantic import BaseModel, Field
10 |
11 | from peatfr_engine.pfvi import simulate_pfvi,
12 |     optimize_pfvi_parameters, calculate_di_obs
13 | from peatfr_engine.imputation import impute_peatfr_data
14 | from peatfr_engine.forecasting import forecast_peatfr_variables
15 | from peatfr_engine.fire_intelligence import fire_engine
16 |
17 | app = FastAPI(
18 |     title="PeatFR EWS Engine API",
19 |     description="Early Warning System & Fire Vulnerability Index
20 |     (PFVI) Forecasting API for Tropical Peatlands",
21 |     version="1.0.0"
22 | )
23 |
24 | app.add_middleware(
25 |     CORSMiddleware,
26 |     allow_origins=["*"],
27 |     allow_credentials=True,
28 |     allow_methods=["*"],
29 |     allow_headers=["*"],
30 | )
31 |
32 | # Load Indonesia Administrative Regions
33 | ADMIN_FILE = os.path.join(os.path.dirname(__file__),
34 |     "peatfr_engine", "indonesia_admin.json")
35 | try:
36 |     with open(ADMIN_FILE) as f:
37 |         INDONESIA_REGIONS = json.load(f)
38 | except Exception:
39 |     INDONESIA_REGIONS = []
40 |
41 | class ImputeRequest(BaseModel):
42 |     WT: List[Optional[float]]
43 |     SM: List[Optional[float]]
44 |     Rf: List[Optional[float]]
45 |     Temp: List[Optional[float]]
46 |     method: str = Field("knn", description="imputation method: knn,
47 |     spline, loess, linear")
48 |     k: int = Field(5, description="k parameter for knn")
49 |
50 | class ForecastRequest(BaseModel):
51 |     WT: List[float]
52 |     SM: List[float]
53 |     Rf: List[float]
54 |     Temp: List[float]
55 |     h: int = Field(4, description="forecasting horizon (days)")
56 |     model: str = Field("arima", description="model type: arima,
57 |     lstm, gru")
58 |     look_back: int = Field(12, description="sliding window for
59 |     LSTM/GRU")
60 |     hidden_units: int = Field(32, description="neurons count")
61 |     epochs: int = Field(50, description="training epochs")
62 |
63 | class AutoPipelineRequest(BaseModel):
64 |     WT: List[Optional[float]]
65 |     SM: List[Optional[float]]
66 |     Rf: List[Optional[float]]

```

```

61 |     Temp: List[Optional[float]]
62 |     imputation: str = Field("knn")
63 |     model: str = Field("arima")
64 |     h: int = Field(4)
65 |     r0: float = Field(3000.0)
66 |     look_back: int = Field(12)
67 |     hidden_units: int = Field(32)
68 |     epochs: int = Field(50)
69 |
70 | @app.get("/")
71 | @app.head("/")
72 | def read_root():
73 |     return {
74 |         "status": "online",
75 |         "service": "PeatFR Early Warning System Engine",
76 |         "version": "1.0.0",
77 |         "docs": "/docs"
78 |     }
79 |
80 | @app.get("/api/v1/health")
81 | @app.head("/api/v1/health")
82 | def health_check():
83 |     return {"status": "ok", "engine": "Python PyTorch & SciPy PeatFR
    Core"}
84 |
85 | @app.get("/api/v1/indonesia/regions")
86 | def get_indonesia_regions():
87 |     """Returns administrative hierarchy of Indonesia Peatland
    Provinces & Regencies."""
88 |     return {"status": "success", "provinces": INDONESIA_REGIONS}
89 |
90 | @app.get("/api/v1/realtime-peatland-data")
91 | def get_realtime_openmeteo_data(
92 |     lat: float = Query(-2.321, description="Latitude coordinate"),
93 |     lon: float = Query(113.901, description="Longitude coordinate"),
94 |     days: int = Query(30, description="Past days of historical open
    satellite data")
95 | ):
96 |     try:
97 |         url = (
98 |             f"https://api.open-meteo.com/v1/forecast?"
99 |             f"latitude={lat}&longitude={lon}&"
100 |             f"daily=temperature_2m_max,precipitation_sum&"
101 |             f"hourly=soil_moisture_0_to_7cm,soil_moisture_7_to_28cm&"
102 |             f"timezone=Asia%2FJakarta&past_days={days}&forecast_days=1"
103 |         )
104 |         req = urllib.request.Request(url, headers={"User-Agent":
    "Mozilla/5.0 (PeatFR-EWS/1.0)"})
105 |         res = urllib.request.urlopen(req, timeout=10)
106 |         data = json.loads(res.read().decode())
107 |
108 |         daily = data["daily"]
109 |         hourly = data["hourly"]
110 |
111 |         dates = daily["time"]
112 |         temp_max = daily["temperature_2m_max"]
113 |         precip = daily["precipitation_sum"]
114 |
115 |         sm_0_7 = np.array(hourly["soil_moisture_0_to_7cm"])
116 |         sm_7_28 = np.array(hourly["soil_moisture_7_to_28cm"])
117 |         sm_combined = (sm_0_7 * 0.4 + sm_7_28 * 0.6)
118 |
119 |         n_days = len(dates)
120 |         sm_raw = sm_combined[:n_days * 24].reshape(n_days,
    24).mean(axis=1)
121 |
122 |         sm_pct = np.clip(32.0 + ((sm_raw - 0.08) / (0.42 - 0.08)) *
    43.0, 30.0, 75.0)
123 |         wt_est = -0.30 - 0.012 * (65.0 - sm_pct) - 0.004 *
    np.maximum(0, 5.0 - np.array(precip))
124 |         wt_est = np.clip(wt_est, -1.8, -0.1)
125 |
126 |         # Multi-Source Fire Intelligence
127 |         fire_nearby = fire_engine.fetch_fireping_nearby(lat, lon,
    radius_m=25000)
128 |         latest_temp = float(temp_max[-1]) if temp_max else 34.0
129 |         latest_rf = float(precip[-1]) if precip else 0.0
130 |         latest_sm = float(sm_pct[-1]) if len(sm_pct) > 0 else 40.0
131 |         fwi_intel = fire_engine.fetch_owm_fwi(lat, lon,
    temp=latest_temp, rf=latest_rf, sm=latest_sm)
132 |
133 |         return {
134 |             "source": "Open-Meteo / ERA5-Land Realtime Satellite
    Data",

```

```

135 |         "lat": lat,
136 |         "lon": lon,
137 |         "dates": dates,
138 |         "WT": np.round(wt_est, 3).tolist(),
139 |         "SM": np.round(sm_pct, 2).tolist(),
140 |         "Rf": np.round(precip, 2).tolist(),
141 |         "Temp": np.round(temp_max, 2).tolist(),
142 |         "fire_intelligence": {
143 |             "nearby": fire_nearby,
144 |             "fwi": fwi_intel
145 |         }
146 |     }
147 | except Exception as e:
148 |     raise HTTPException(status_code=500, detail=f"Failed to
149 |     fetch realtime open data: {str(e)}")
150 | @app.get("/api/v1/fire-intelligence/nearby")
151 | def get_nearby_fires(
152 |     lat: float = Query(-2.321, description="Latitude"),
153 |     lon: float = Query(113.901, description="Longitude"),
154 |     radius_m: int = Query(25000, description="Radius in meters")
155 | ):
156 |     """Fetch nearby satellite fire detection & GWIS 7-day burned
157 |     area (FirePing Public API)."""
158 |     return fire_engine.fetch_fireping_nearby(lat, lon,
159 |     radius_m=radius_m)
160 | @app.get("/api/v1/fire-intelligence/gfw-peatland-fires")
161 | def get_gfw_peatland_fires(
162 |     limit: int = Query(100, description="Maximum fire alerts")
163 | ):
164 |     """Fetch vector NASA VIIRS active fire alerts specifically on
165 |     Indonesian peatlands (is_peatland = true)."""
166 |     return fire_engine.fetch_gfw_peatland_fires(limit=limit)
167 | @app.get("/api/v1/fire-intelligence/firms-hotspots")
168 | def get_firms_hotspots(
169 |     bbox: str = Query("95,-11,141,6", description="Bounding box
170 |     [min_lon,min_lat,max_lon,max_lat]"),
171 |     source: str = Query("VIIRS_SNPP_NRT", description="Satellite
172 |     sensor: VIIRS_SNPP_NRT, MODIS_NRT, VIIRS_NOAA20_NRT,
173 |     VIIRS_NOAA21_NRT"),
174 |     day_range: int = Query(1, ge=1, le=10, description="Range of
175 |     days back (1-10)")
176 | ):
177 |     """Fetch live NASA FIRMS active fire hotspots via authorized
178 |     MAP_KEY."""
179 |     return fire_engine.fetch_firms_hotspots(bbox=bbox,
180 |     source=source, day_range=day_range)
181 | @app.get("/api/v1/fire-intelligence/severe-alerts")
182 | def get_severe_fire_alerts():
183 |     """Fetch active severe fire alerts (Kebakaran Parah) across
184 |     Indonesian Peatlands."""
185 |     return fire_engine.fetch_severe_fire_alerts()
186 | @app.get("/api/v1/fire-intelligence/fwi")
187 | def get_fwi_forecast(
188 |     lat: float = Query(-2.321, description="Latitude"),
189 |     lon: float = Query(113.901, description="Longitude")
190 | ):
191 |     """Fetch Fire Weather Index (FWI) forecast (OpenWeatherMap /
192 |     Canadian FWI Engine)."""
193 |     return fire_engine.fetch_owm_fwi(lat, lon)
194 | @app.get("/api/v1/sabangau-sample")
195 | def get_sabangau_sample():
196 |     np.random.seed(42)
197 |     days = 192
198 |     time_idx = np.arange(days)
199 |
200 |     wt = -0.4 - 0.65 * (1.0 / (1.0 + np.exp(-(time_idx - 100) /
201 |     20.0))) + np.random.normal(0, 0.02, days)
202 |     sm = 65.0 - 27.0 * (1.0 / (1.0 + np.exp(-(time_idx - 100) /
203 |     20.0))) + np.random.normal(0, 1.2, days)
204 |     rf = np.maximum(0.0, np.random.exponential(8.0, days) * (1.0 -
205 |     (time_idx / 220.0)))
206 |     temp = 31.5 + 4.0 * (1.0 / (1.0 + np.exp(-(time_idx - 100) /
207 |     25.0))) + np.random.normal(0, 0.4, days)
208 |
209 |     wt_raw = wt.copy()
210 |     sm_raw = sm.copy()
211 |     rf_raw = rf.copy()
212 |     temp_raw = temp.copy()

```

```

203 |
204 |     wt_raw[114:117] = np.nan
205 |     sm_raw[114:117] = np.nan
206 |     rf_raw[114:117] = np.nan
207 |     temp_raw[145] = np.nan
208 |
209 |     return {
210 |         "location": "Sabangau, Central Kalimantan, Indonesia",
211 |         "time_steps": days,
212 |         "WT": [None if np.isnan(x) else round(float(x), 3) for x in
wt_raw],
213 |         "SM": [None if np.isnan(x) else round(float(x), 2) for x in
sm_raw],
214 |         "Rf": [None if np.isnan(x) else round(float(x), 2) for x in
rf_raw],
215 |         "Temp": [None if np.isnan(x) else round(float(x), 2) for x
in temp_raw]
216 |     }
217 |
218 | @app.post("/api/v1/impute")
219 | def impute_data(req: ImputeRequest):
220 |     try:
221 |         wt = np.array(req.WT, dtype=float)
222 |         sm = np.array(req.SM, dtype=float)
223 |         rf = np.array(req.Rf, dtype=float)
224 |         temp = np.array(req.Temp, dtype=float)
225 |
226 |         res = impute_peatfr_data(wt, sm, rf, temp,
method=req.method, k=req.k)
227 |
228 |         return {
229 |             "method": req.method,
230 |             "WT": res["WT"].tolist(),
231 |             "SM": res["SM"].tolist(),
232 |             "Rf": res["Rf"].tolist(),
233 |             "Temp": res["Temp"].tolist()
234 |         }
235 |     except Exception as e:
236 |         raise HTTPException(status_code=400, detail=str(e))
237 |
238 | @app.post("/api/v1/forecast")
239 | def forecast_data(req: ForecastRequest):
240 |     try:
241 |         wt = np.array(req.WT, dtype=float)
242 |         sm = np.array(req.SM, dtype=float)
243 |         rf = np.array(req.Rf, dtype=float)
244 |         temp = np.array(req.Temp, dtype=float)
245 |
246 |         res = forecast_peatfr_variables(
247 |             wt, sm, rf, temp,
248 |             h=req.h,
249 |             model=req.model,
250 |             look_back=req.look_back,
251 |             hidden_units=req.hidden_units,
252 |             epochs=req.epochs
253 |         )
254 |
255 |         return {
256 |             "model": req.model,
257 |             "h": req.h,
258 |             "WT_pred": res["WT_pred"].tolist(),
259 |             "SM_pred": res["SM_pred"].tolist(),
260 |             "Rf_pred": res["Rf_pred"].tolist(),
261 |             "Temp_pred": res["Temp_pred"].tolist()
262 |         }
263 |     except Exception as e:
264 |         raise HTTPException(status_code=400, detail=str(e))
265 |
266 | @app.post("/api/v1/pipeline/auto")
267 | def run_auto_pipeline(req: AutoPipelineRequest):
268 |     try:
269 |         wt_in = np.array(req.WT, dtype=float)
270 |         sm_in = np.array(req.SM, dtype=float)
271 |         rf_in = np.array(req.Rf, dtype=float)
272 |         temp_in = np.array(req.Temp, dtype=float)
273 |
274 |         has_nans = any(np.isnan(wt_in)) or any(np.isnan(sm_in)) or
any(np.isnan(rf_in)) or any(np.isnan(temp_in))
275 |
276 |         if has_nans:
277 |             imp_res = impute_peatfr_data(wt_in, sm_in, rf_in,
temp_in, method=req.imputation)
278 |             wt_clean = imp_res["WT"]
279 |             sm_clean = imp_res["SM"]

```

```

280 |         rf_clean = imp_res["Rf"]
281 |         temp_clean = imp_res["Temp"]
282 |     else:
283 |         wt_clean, sm_clean, rf_clean, temp_clean = wt_in, sm_in,
rf_in, temp_in
284 |
285 |         fc_res = forecast_peatfr_variables(
286 |             wt_clean, sm_clean, rf_clean, temp_clean,
287 |             h=req.h,
288 |             model=req.model,
289 |             look_back=req.look_back,
290 |             hidden_units=req.hidden_units,
291 |             epochs=req.epochs
292 |         )
293 |
294 |         wt_full = np.concatenate([wt_clean, fc_res["WT_pred"]])
295 |         sm_full = np.concatenate([sm_clean, fc_res["SM_pred"]])
296 |         rf_full = np.concatenate([rf_clean, fc_res["Rf_pred"]])
297 |         temp_full = np.concatenate([temp_clean,
fc_res["Temp_pred"]])
298 |
299 |         opt_params, opt_mse = optimize_pfvi_parameters(wt_clean,
sm_clean, rf_clean, temp_clean, r0=req.r0)
300 |         a_h, b_h, n, alpha = opt_params
301 |
302 |         pfvi_sim = simulate_pfvi(wt_full, sm_full, rf_full,
temp_full, a_h, b_h, n, alpha, r0=req.r0)
303 |
304 |         pfvi_series = pfvi_sim["pfvi"].tolist()
305 |         pfvi_forecast = pfvi_series[-req.h:]
306 |
307 |         risk_categories = []
308 |         for val in pfvi_forecast:
309 |             if val < 75:
310 |                 risk_categories.append("Low")
311 |             elif val < 150:
312 |                 risk_categories.append("Moderate")
313 |             elif val < 225:
314 |                 risk_categories.append("High")
315 |             else:
316 |                 risk_categories.append("Extreme")
317 |
318 |         current_status = risk_categories[-1] if risk_categories else
"Unknown"
319 |
320 |         return {
321 |             "status": "success",
322 |             "has_missing_data_imputed": has_nans,
323 |             "imputation_method": req.imputation,
324 |             "forecasting_model": req.model,
325 |             "optimization": {
326 |                 "a_h": float(a_h),
327 |                 "b_h": float(b_h),
328 |                 "n": float(n),
329 |                 "alpha": float(alpha),
330 |                 "mse": float(opt_mse)
331 |             },
332 |             "historical_len": len(wt_clean),
333 |             "forecast_horizon": req.h,
334 |             "forecast": {
335 |                 "WT": fc_res["WT_pred"].tolist(),
336 |                 "SM": fc_res["SM_pred"].tolist(),
337 |                 "Rf": fc_res["Rf_pred"].tolist(),
338 |                 "Temp": fc_res["Temp_pred"].tolist(),
339 |                 "PFVI": pfvi_forecast,
340 |                 "Risk_Categories": risk_categories,
341 |                 "Current_Status": current_status
342 |             },
343 |             "full_series": {
344 |                 "WT": wt_full.tolist(),
345 |                 "SM": sm_full.tolist(),
346 |                 "Rf": rf_full.tolist(),
347 |                 "Temp": temp_full.tolist(),
348 |                 "PFVI": pfvi_series,
349 |                 "DI_obs": pfvi_sim["di_obs"].tolist(),
350 |                 "DF": pfvi_sim["df"].tolist(),
351 |                 "RF": pfvi_sim["rf"].tolist(),
352 |                 "WTF": pfvi_sim["wtf"].tolist()
353 |             }
354 |         }
355 |     except Exception as e:
356 |         raise HTTPException(status_code=400, detail=str(e))

```

Screenshot Aplikasi:*[Screenshot: FastAPI Backend: routing API & pipeline auto (dokumentasi /docs) — ss_api.png]***FITUR 9: PeatFR Mathematical Engine & Optimasi Nelder-Mead (Physical Core)**

Deskripsi & Tujuan Fitur: Inti matematis aplikasi: fungsi `calculate_df` (laju evapotranspirasi dari suhu maksimum & PFVI sebelumnya), `calculate_rf` (curah hujan efektif dengan interception 5.1 mm/hari), `calculate_wtf` (faktor muka air tanah menggunakan kurva retensi van Genuchten $\theta(v)$), `calculate_di_obs` (drought index observasi dari soil moisture, dengan $fc=40\%$ / $sat=70\%$ untuk gambut), `simulate_pfvi` (simulasi rekursif PFVI 0–300 dengan konstrain fisik), dan `optimize_pfvi_parameters` (grid search awal multi-titik + minimasi Nelder-Mead terhadap MSE PFVI vs DI_obs).

Cara Menggunakan (Peneliti):

1. Pahami persamaan fisika pada Bab 3 — DF menaikkan PFVI (kekeringan), RF menurunkan (hujan), WTF menurunkan (kapilaritas tanah basah).
1. Optimasi dilakukan otomatis pada setiap pipeline; hasil (a_H , b_H , n , α , MSE) tampil pada panel Nelder-Mead.
1. Ubah tipe lahan (gambut vs mineral) dengan memvariasikan fc/sat pada pemanggilan ``calculate_di_obs``.

Kode Sumber (Source Code Listing): backend/peatfr_engine/pfvi.py

```

1 | import numpy as np
2 | from scipy.optimize import minimize
3 | from typing import Dict, Any, Tuple, List
4 |
5 | def calculate_df(pfvi_prev: float, temp_max: float, r0: float =
6 |     3000.0, dt: float = 1.0) -> float:
7 |     """
8 |     Evapotranspiration water loss DF_t equation from Mahdiyasa et
9 |     al. (2025).
10 |     """
11 |     numerator = (300.0 - pfvi_prev) * (0.4982 * np.exp(0.0905 *
12 |         temp_max + 1.6096) - 4.268) * dt * 1e-3
13 |     denominator = 1.0 + 10.88 * np.exp(-0.00173582677165354 * r0)
14 |     return float(numerator / denominator)
15 |
16 | def calculate_rf(rf_current: float, rf_prev: float) -> float:
17 |     """
18 |     Effective rainfall factor RF_t accounting for initial
19 |     canopy/surface interception (5.1 mm/day).
20 |     """
21 |     if rf_prev is None or np.isnan(rf_prev) or rf_prev <= 5.1:
22 |         if rf_current < 5.1:
23 |             return 0.0
24 |         else:
25 |             return float(rf_current - 5.1)
26 |     else:
27 |         if rf_current >= 5.1:
28 |             return float(rf_current)
29 |         else:
30 |             return 0.0
31 |
32 | def calculate_wtf(wt_depth: float, a_h: float, b_h: float, n: float,
33 |     alpha: float) -> Tuple[float, float]:
34 |     """
35 |     Water table factor WTF_t using van Genuchten soil moisture
36 |     retention function theta(v).
37 |     wt_depth: positive depth from surface to water table in cm.
38 |     """
39 |     if n <= 0 or alpha <= 0:
40 |         return 0.0, 0.0
41 |
42 |     m = 1.0 - (1.0 / n)
43 |     v = max(0.0, wt_depth)
44 |     theta = (1.0 + (v / alpha) ** n) ** (-m)
45 |     wtf = a_h - b_h * ((1.0 - theta) * 300.0)
46 |     return float(wtf), float(theta)

```

```

42 | def calculate_di_obs(sm: np.ndarray, fc: float = 40.0, sat: float =
43 |     70.0) -> np.ndarray:
44 |     """
45 |     Observed drought index derived from volumetric soil moisture
46 |     (SM).
47 |     """
48 |     di = 300.0 * (1.0 - ((sm - fc) / (sat - fc)))
49 |     return np.clip(di, 0.0, 300.0)
50 |
51 | def simulate_pfvi(
52 |     wt: np.ndarray,
53 |     sm: np.ndarray,
54 |     rf: np.ndarray,
55 |     temp: np.ndarray,
56 |     a_h: float,
57 |     b_h: float,
58 |     n: float,
59 |     alpha: float,
60 |     r0: float = 3000.0,
61 |     dt: float = 1.0
62 | ) -> Dict[str, Any]:
63 |     """
64 |     Simulates time series of Peat Fire Vulnerability Index (PFVI).
65 |     """
66 |     time_steps = len(wt)
67 |     depths = np.where(wt > 0, 0.0, -wt) * 100.0 # convert meters to
68 |     cm if negative meters
69 |
70 |     pfvi = np.zeros(time_steps)
71 |     df_arr = np.zeros(time_steps)
72 |     rf_arr = np.zeros(time_steps)
73 |     wtf_arr = np.zeros(time_steps)
74 |     theta_arr = np.zeros(time_steps)
75 |
76 |     di_obs = calculate_di_obs(sm)
77 |     current_pfvi = float(di_obs[0])
78 |
79 |     for t in range(time_steps):
80 |         pfvi_clamped = max(0.0, min(300.0, current_pfvi))
81 |         rf_prev = rf[t-1] if t > 0 else None
82 |
83 |         df_val = calculate_df(pfvi_clamped, temp[t], r0, dt)
84 |         rf_val = calculate_rf(rf[t], rf_prev)
85 |         wtf_val, theta_val = calculate_wtf(depths[t], a_h, b_h, n,
86 |         alpha)
87 |
88 |         next_pfvi = pfvi_clamped + df_val - rf_val - wtf_val
89 |         next_pfvi = max(0.0, min(300.0, next_pfvi))
90 |
91 |         pfvi[t] = next_pfvi
92 |         df_arr[t] = df_val
93 |         rf_arr[t] = rf_val
94 |         wtf_arr[t] = wtf_val
95 |         theta_arr[t] = theta_val
96 |         current_pfvi = next_pfvi
97 |
98 |     return {
99 |         "pfvi": pfvi,
100 |         "di_obs": di_obs,
101 |         "df": df_arr,
102 |         "rf": rf_arr,
103 |         "wtf": wtf_arr,
104 |         "theta": theta_arr
105 |     }
106 |
107 | def optimize_pfvi_parameters(
108 |     wt: np.ndarray,
109 |     sm: np.ndarray,
110 |     rf: np.ndarray,
111 |     temp: np.ndarray,
112 |     r0: float = 3000.0,
113 |     dt: float = 1.0
114 | ) -> Tuple[np.ndarray, float]:
115 |     """
116 |     Calibrates (a_h, b_h, n, alpha) using Nelder-Mead optimization
117 |     against observed DI.
118 |     """
119 |     di_obs = calculate_di_obs(sm)
120 |
121 |     def objective(params):
122 |         a_h, b_h, n, alpha = params
123 |         if n <= 0 or alpha <= 0:
124 |             return 1e9
125 |         sim = simulate_pfvi(wt, sm, rf, temp, a_h, b_h, n, alpha,

```

```

121 |     r0, dt)
122 |     mse = np.mean((sim["pfvi"] - di_obs) ** 2)
123 |     return mse
124 |
125 |     # Initial grid search for robust starting point
126 |     best_params = np.array([0.1, 0.1, 0.1, 0.1])
127 |     best_val = objective(best_params)
128 |
129 |     for i in [0.2, 0.5, 1.0, 5.0]:
130 |         for j in [0.01, 0.05, 0.1, 0.5]:
131 |             for k in [1.5, 5.0, 15.0, 55.0]:
132 |                 for l in [0.5, 1.0, 5.0]:
133 |                     init_p = np.array([i, j, k, l])
134 |                     val = objective(init_p)
135 |                     if val < best_val:
136 |                         best_val = val
137 |                         best_params = init_p
138 |
139 |     res = minimize(objective, best_params, method='Nelder-Mead',
140 |                   options={'maxiter': 2000, 'xatol': 1e-4, 'fatol': 1e-4})
141 |     return res.x, float(res.fun)

```

FITUR 10: Imputation Engine Data Missing (kNN, Spline, LOESS, Linear)

Deskripsi & Tujuan Fitur: Modul `imputation.py` menyediakan empat algoritma pengisian data kosong (NaN) pada deret time-series telemetri: **linear** (interpolasi pandas limit_direction='both'), **spline** (CubicSpline dengan ekstrapolasi; fallback linear jika < 4 titik valid), **loess** (LOWESS statsmodels dengan interpolasi; fallback linear jika < 10 titik valid), dan **kNN** (imputasi multivariat berbasis jarak Gower-like pada data ternormalisasi [0,1], rata-rata k tetangga terdekat + fallback interpolasi linear).

Cara Menggunakan (Peneliti):

1. Pipeline otomatis mendeteksi NaN dan memanggil metode terpilih (default kNN).
1. Pengujian mandiri: ``POST /api/v1/impute`` dengan array yang mengandung ``null`` pada WT/SM/Rf/Temp.
1. Dataset sampel ``/api/v1/sabangau-sample`` sengaja mengandung NaN pada indeks 114–117 dan 145 untuk uji imputasi.

Kode Sumber (Source Code Listing): backend/peatfr_engine/imputation.py

```

1 | import numpy as np
2 | import pandas as pd
3 | from scipy.interpolate import interp1d, CubicSpline
4 | from typing import Dict, Any, List
5 |
6 | def linear_impute(arr: np.ndarray) -> np.ndarray:
7 |     """Linear interpolation for 1D numpy array."""
8 |     series = pd.Series(arr)
9 |     imputed = series.interpolate(method='linear',
10 |                                limit_direction='both')
11 |     return imputed.to_numpy()
12 |
13 | def spline_impute(arr: np.ndarray) -> np.ndarray:
14 |     """Cubic spline interpolation for 1D numpy array."""
15 |     valid_idx = np.where(~np.isnan(arr))[0]
16 |     if len(valid_idx) < 4:
17 |         return linear_impute(arr)
18 |     cs = CubicSpline(valid_idx, arr[valid_idx], extrapolate=True)
19 |     all_idx = np.arange(len(arr))
20 |     return cs(all_idx)
21 |
22 | def loess_impute(arr: np.ndarray, span: float = 0.5) -> np.ndarray:
23 |     """LOESS / LOWESS local regression smoothing for imputation."""
24 |     valid_idx = np.where(~np.isnan(arr))[0]
25 |     if len(valid_idx) < 10:
26 |         return linear_impute(arr)
27 |     try:
28 |         from statsmodels.nonparametric.smoothers_lowess import
29 |         lowess
30 |         res = lowess(arr[valid_idx], valid_idx, frac=span,
31 |                      is_sorted=True, return_sorted=False)
32 |         f = interp1d(valid_idx, res, kind='linear',

```

```

    fill_value='extrapolate')
30 |     all_idx = np.arange(len(arr))
31 |     return f(all_idx)
32 | except Exception:
33 |     return linear_impute(arr)
34 |
35 | def knn_impute_dataset(df: pd.DataFrame, k: int = 5) ->
    pd.DataFrame:
36 |     """
37 |     kNN Imputation using Gower-like normalized Euclidean distance on
    multivariate time series.
38 |     """
39 |     cols = df.columns
40 |     data = df.to_numpy(dtype=float, copy=True)
41 |     n_samples, n_features = data.shape
42 |
43 |     # Normalize features to [0, 1] for Gower-like scaling
44 |     mins = np.nanmin(data, axis=0)
45 |     maxs = np.nanmax(data, axis=0)
46 |     ranges = np.where((maxs - mins) == 0, 1.0, maxs - mins)
47 |     norm_data = (data - mins) / ranges
48 |
49 |     for i in range(n_samples):
50 |         if np.isnan(data[i]).any():
51 |             missing_cols = np.where(np.isnan(data[i]))[0]
52 |             observed_cols = np.where(~np.isnan(data[i]))[0]
53 |
54 |             if len(observed_cols) == 0:
55 |                 continue
56 |
57 |             # Compute distance to other samples based on observed
    columns
58 |             distances = []
59 |             for j in range(n_samples):
60 |                 if i == j or np.isnan(data[j]).any():
61 |                     distances.append((1e9, j))
62 |                 else:
63 |                     dist = np.mean(np.abs(norm_data[i,
    observed_cols] - norm_data[j, observed_cols]))
64 |                     distances.append((dist, j))
65 |
66 |             distances.sort(key=lambda x: x[0])
67 |             neighbors_idx = [idx for dist, idx in distances[:k] if
    dist < 1e8]
68 |
69 |             if neighbors_idx:
70 |                 for c in missing_cols:
71 |                     neighbor_vals = data[neighbors_idx, c]
72 |                     data[i, c] = np.mean(neighbor_vals)
73 |
74 |             # Final fallback for any remaining NaNs
75 |             res_df = pd.DataFrame(data, columns=cols)
76 |             return res_df.interpolate(method='linear',
    limit_direction='both')
77 |
78 | def impute_peatfr_data(
79 |     wt: np.ndarray,
80 |     sm: np.ndarray,
81 |     rf: np.ndarray,
82 |     temp: np.ndarray,
83 |     method: str = "knn",
84 |     k: int = 5
85 | ) -> Dict[str, np.ndarray]:
86 |     """
87 |     Unified entry point for data imputation in peatfr.
88 |     """
89 |     df = pd.DataFrame({"WT": wt, "SM": sm, "Rf": rf, "Temp": temp})
90 |
91 |     if method == "knn":
92 |         imputed_df = knn_impute_dataset(df, k=k)
93 |     elif method == "spline":
94 |         imputed_df = df.apply(spline_impute)
95 |     elif method == "loess":
96 |         imputed_df = df.apply(loess_impute)
97 |     else:
98 |         imputed_df = df.apply(linear_impute)
99 |
100 |     return {
101 |         "WT": imputed_df["WT"].to_numpy(),
102 |         "SM": imputed_df["SM"].to_numpy(),
103 |         "Rf": imputed_df["Rf"].to_numpy(),
104 |         "Temp": imputed_df["Temp"].to_numpy()
105 |     }

```

FITUR 11: Forecasting Engine (ARIMA + Box-Cox & PyTorch LSTM/GRU)

Deskripsi & Tujuan Fitur: Modul `forecasting.py` mengimplementasikan tiga model: **ARIMA** (transformasi Box-Cox dengan shift positif bila ada nilai ≤ 0 , pencarian AIC atas $p \in \{1,2\}$, $d \in \{0,1\}$, $q \in \{0,1,2\}$, inverse Box-Cox, fallback tren linear), **LSTM** dan **GRU** (jaringan saraf PyTorch 2-layer dengan `hidden_units`, `look_back` sliding window, training Adam $lr=0.01$ & `MSELoss`, prediksi autoregresif iteratif dengan `update window`). Fungsi `forecast_peatfr_variables` menjalankan keempat variabel (WT/SM/Rf/Temp) dengan model yang sama.

Cara Menggunakan (Peneliti):

1. Pilih model di panel Pipeline Controls atau via ``POST /api/v1/forecast`` (field model: `arima|lstm|gru`).
1. Ubah ``look_back`` (window), ``hidden_units``, ``epochs`` untuk tuning LSTM/GRU.
1. ARIMA dipilih default karena cepat & deterministik; LSTM/GRU direkomendasikan untuk dataset panjang (>30 titik).

Kode Sumber (Source Code Listing): `backend/peatfr_engine/forecasting.py`

```

1 | import numpy as np
2 | import pandas as pd
3 | from typing import Dict, Any, Tuple, List
4 | from scipy.stats import boxcox
5 | from scipy.special import inv_boxcox
6 |
7 | # Machine learning PyTorch imports
8 | import torch
9 | import torch.nn as nn
10 |
11 | class LSTMModel(nn.Module):
12 |     def __init__(self, input_dim: int = 1, hidden_dim: int = 32,
13 |                  num_layers: int = 2, output_dim: int = 1):
14 |         super(LSTMModel, self).__init__()
15 |         self.lstm = nn.LSTM(input_dim, hidden_dim, num_layers,
16 |                              batch_first=True)
17 |         self.fc = nn.Linear(hidden_dim, output_dim)
18 |
19 |     def forward(self, x):
20 |         out, _ = self.lstm(x)
21 |         out = self.fc(out[:, -1, :])
22 |         return out
23 |
24 | class GRUModel(nn.Module):
25 |     def __init__(self, input_dim: int = 1, hidden_dim: int = 32,
26 |                  num_layers: int = 2, output_dim: int = 1):
27 |         super(GRUModel, self).__init__()
28 |         self.gru = nn.GRU(input_dim, hidden_dim, num_layers,
29 |                             batch_first=True)
30 |         self.fc = nn.Linear(hidden_dim, output_dim)
31 |
32 |     def forward(self, x):
33 |         out, _ = self.gru(x)
34 |         out = self.fc(out[:, -1, :])
35 |         return out
36 |
37 | def forecast_arima_single(series: np.ndarray, h: int) -> np.ndarray:
38 |     """
39 |     ARIMA forecasting with Box-Cox transformation & auto AR/MA
40 |     fitting.
41 |     """
42 |     shift = 0.0
43 |     min_val = np.min(series)
44 |     if min_val <= 0:
45 |         shift = abs(min_val) + 1.0
46 |
47 |     shifted_series = series + shift
48 |
49 |     try:
50 |         transformed, lambda = boxcox(shifted_series)
51 |     except Exception:
52 |         transformed = np.log(shifted_series)
53 |         lambda = 0.0
54 |
55 |     try:
56 |         from statsmodels.tsa.arima.model import ARIMA

```

```

52 |         # Fit optimal ARIMA(p, d, q) based on AIC
53 |         best_aic = 1e9
54 |         best_model = None
55 |
56 |         for p in [1, 2]:
57 |             for d in [0, 1]:
58 |                 for q in [0, 1, 2]:
59 |                     try:
60 |                         m = ARIMA(transformed, order=(p, d,
61 | q)).fit()
62 |                         if m.aic < best_aic:
63 |                             best_aic = m.aic
64 |                             best_model = m
65 |                     except Exception:
66 |                         continue
67 |
68 |         if best_model is None:
69 |             best_model = ARIMA(transformed, order=(1, 1, 1)).fit()
70 |
71 |         forecast_trans = best_model.forecast(steps=h)
72 |
73 |         if lambda == 0.0:
74 |             forecast_shifted = np.exp(forecast_trans)
75 |         else:
76 |             forecast_shifted = inv_boxcox(forecast_trans, lambda)
77 |
78 |         forecast = forecast_shifted - shift
79 |         return np.array(forecast)
80 |     except Exception:
81 |         # Fallback to linear trend if ARIMA fails
82 |         last_val = series[-1]
83 |         trend = (series[-1] - series[0]) / len(series)
84 |         return np.array([last_val + trend * (i + 1) for i in
85 | range(h)])
86 |
87 | def forecast_rnn_single(series: np.ndarray, h: int, model_type: str
88 | = "lstm", look_back: int = 12, hidden_units: int = 32, epochs: int =
89 | 50) -> np.ndarray:
90 |     """
91 |     LSTM/GRU neural network forecasting for univariate time series.
92 |     """
93 |     data = series.astype(np.float32)
94 |     min_val, max_val = data.min(), data.max()
95 |     scale = (max_val - min_val) if (max_val - min_val) != 0 else 1.0
96 |     norm_data = (data - min_val) / scale
97 |
98 |     if len(norm_data) <= look_back:
99 |         look_back = max(2, len(norm_data) // 2)
100 |
101 |     X, y = [], []
102 |     for i in range(len(norm_data) - look_back):
103 |         X.append(norm_data[i:i + look_back])
104 |         y.append(norm_data[i + look_back])
105 |
106 |     X_tensor = torch.tensor(np.array(X),
107 | dtype=torch.float32).unsqueeze(-1)
108 |     y_tensor = torch.tensor(np.array(y),
109 | dtype=torch.float32).unsqueeze(-1)
110 |
111 |     if model_type == "gru":
112 |         model = GRUModel(input_dim=1, hidden_dim=hidden_units,
113 | num_layers=2, output_dim=1)
114 |     else:
115 |         model = LSTMModel(input_dim=1, hidden_dim=hidden_units,
116 | num_layers=2, output_dim=1)
117 |
118 |     optimizer = torch.optim.Adam(model.parameters(), lr=0.01)
119 |     criterion = nn.MSELoss()
120 |
121 |     model.train()
122 |     for epoch in range(epochs):
123 |         optimizer.zero_grad()
124 |         output = model(X_tensor)
125 |         loss = criterion(output, y_tensor)
126 |         loss.backward()
127 |         optimizer.step()
128 |
129 |     model.eval()
130 |     preds = []
131 |     curr_input = torch.tensor(norm_data[-look_back:],
132 | dtype=torch.float32).unsqueeze(0).unsqueeze(-1)
133 |
134 |     with torch.no_grad():
135 |         for _ in range(h):

```

```

127 |         next_pred = model(curr_input)
128 |         preds.append(next_pred.item())
129 |         # Update sliding window
130 |         next_val_tensor = next_pred.unsqueeze(1)
131 |         curr_input = torch.cat((curr_input[:, 1:, :],
132 |                                next_val_tensor), dim=1)
133 |     preds = np.array(preds) * scale + min_val
134 |     return preds
135 |
136 | def forecast_peatfr_variables(
137 |     wt: np.ndarray,
138 |     sm: np.ndarray,
139 |     rf: np.ndarray,
140 |     temp: np.ndarray,
141 |     h: int = 4,
142 |     model: str = "arima",
143 |     look_back: int = 12,
144 |     hidden_units: int = 32,
145 |     epochs: int = 50
146 | ) -> Dict[str, np.ndarray]:
147 |     """
148 |     Unified entry point for time series forecasting in peatfr.
149 |     """
150 |     model_name = model.lower()
151 |
152 |     if model_name == "arima":
153 |         wt_pred = forecast_arima_single(wt, h)
154 |         sm_pred = forecast_arima_single(sm, h)
155 |         rf_pred = forecast_arima_single(rf, h)
156 |         temp_pred = forecast_arima_single(temp, h)
157 |     else:
158 |         wt_pred = forecast_rnn_single(wt, h, model_type=model_name,
159 |                                       look_back=look_back, hidden_units=hidden_units, epochs=epochs)
160 |         sm_pred = forecast_rnn_single(sm, h, model_type=model_name,
161 |                                       look_back=look_back, hidden_units=hidden_units, epochs=epochs)
162 |         rf_pred = forecast_rnn_single(rf, h, model_type=model_name,
163 |                                       look_back=look_back, hidden_units=hidden_units, epochs=epochs)
164 |         temp_pred = forecast_rnn_single(temp, h,
165 |                                         model_type=model_name, look_back=look_back,
166 |                                         hidden_units=hidden_units, epochs=epochs)
167 |
168 |     return {
169 |         "WT_pred": wt_pred,
170 |         "SM_pred": sm_pred,
171 |         "Rf_pred": rf_pred,
172 |         "Temp_pred": temp_pred
173 |     }

```

FITUR 12: Multi-Source Fire Intelligence Engine (Satellite Aggregator)

Deskripsi & Tujuan Fitur: Modul `fire_intelligence.py` mengagregasi 5 sumber data kebakaran dalam satu kelas `FireIntelligenceEngine` dengan **SimpleTTLCache 180 detik** (menghemat kuota API NASA): **NASA FIRMS Area API** (CSV hotspot VIIRS/MODIS, parsing header dinamis, MAP_KEY default aa16407e...fe020), **Severe Fire Alerts** (struktur peringatan Katingan/Kotim/Sabangau/Siak dengan estimasi area terbakar), **FirePing Public API** (deteksi terdekat, burned area GWIS), **GFW Data API** (query SQL `iso='IDN' AND is__peatland=true`), dan **OpenWeather FWI** (fallback estimasi meteorologis: $\text{dryness} \times 0.4 + \text{heat} \times 1.2 - \text{rain} \times 2.0$, rating Low→Extreme).

Cara Menggunakan (Developer):

1. Akses ``/api/v1/fire-intelligence/nearby?lat&lon;&radius;_m=25000`` untuk deteksi api terdekat.
1. Akses ``/api/v1/fire-intelligence/firms-hotspots?bbox=95,-11,141,6&source;=VIIRS_SNPP_NRT&day;_range=1`` untuk hotspot nasional.
1. Akses ``/api/v1/fire-intelligence/gfw-peatland-fires?limit=100`` untuk kebakaran spesifik lahan gambut.
1. Akses ``/api/v1/fire-intelligence/severe-alerts`` untuk peringatan parah terstruktur.
1. Nilai MAP_KEY dapat di-override via environment variable ``FIRMS_MAP_KEY`` pada container.

Kode Sumber (Source Code Listing): backend/peatfr_engine/fire_intelligence.py

```

1 | import os
2 | import json
3 | import time
4 | import urllib.request
5 | import urllib.parse
6 | from typing import Dict, Any, Optional
7 |
8 | class SimpleTTLCache:
9 |     def __init__(self, ttl_seconds: int = 180):
10 |         self.ttl = ttl_seconds
11 |         self._cache: Dict[str, tuple[float, Any]] = {}
12 |
13 |     def get(self, key: str) -> Optional[Any]:
14 |         if key in self._cache:
15 |             entry_time, data = self._cache[key]
16 |             if time.time() - entry_time < self.ttl:
17 |                 return data
18 |             else:
19 |                 del self._cache[key]
20 |         return None
21 |
22 |     def set(self, key: str, data: Any):
23 |         self._cache[key] = (time.time(), data)
24 |
25 | class FireIntelligenceEngine:
26 |     def __init__(self, gfw_api_key: Optional[str] = None,
27 |                  owm_api_key: Optional[str] = None, firms_map_key: Optional[str] =
28 |                  None):
29 |         self.gfw_api_key = gfw_api_key or os.getenv("GFW_API_KEY")
30 |         self.owm_api_key = owm_api_key or
31 |         os.getenv("OPENWEATHER_API_KEY")
32 |         self.firms_map_key = firms_map_key or
33 |         os.getenv("FIRMS_MAP_KEY", "aa16407e5eb11df46b09cafc085fe020")
34 |         self.headers = {"User-Agent": "Mozilla/5.0 (PeatFR-EWS/1.0;
35 |         https://peatfr.bagaswibowo.app)"}
36 |         self.cache = SimpleTTLCache(ttl_seconds=180)
37 |
38 |     def fetch_firms_hotspots(self, bbox: str = "95,-11,141,6",
39 |                             source: str = "VIIRS_SNPP_NRT", day_range: int = 1) -> Dict[str,
40 |                             Any]:
41 |         """
42 |         Fetch real-time active fire hotspots directly from NASA
43 |         FIRMS Area API using MAP_KEY.
44 |         Uses in-memory 180s TTL cache to optimize performance & NASA
45 |         transaction quota.
46 |         """
47 |         cache_key = f"firms:{source}:{bbox}:{day_range}"
48 |         cached_res = self.cache.get(cache_key)
49 |         if cached_res:
50 |             return cached_res
51 |
52 |         url = f"https://firms.modaps.eosdis.nasa.gov/api/area/csv/{s
53 |         elf.firms_map_key}/{source}/{bbox}/{day_range}"
54 |         req = urllib.request.Request(url, headers=self.headers)
55 |         try:
56 |             with urllib.request.urlopen(req, timeout=12) as resp:
57 |                 csv_text = resp.read().decode('utf-8').strip()
58 |                 lines = csv_text.splitlines()
59 |                 if not lines or lines[0].startswith("Invalid") or
60 |                 lines[0].startswith("Error"):
61 |                     return {"status": "error", "message": csv_text,
62 |                     "count": 0, "hotspots": []}
63 |
64 |                 header = [h.strip() for h in lines[0].split(",")]
65 |                 hotspots = []
66 |                 for line in lines[1:]:
67 |                     parts = [p.strip() for p in line.split(",")]
68 |                     if len(parts) == len(header):
69 |                         row = dict(zip(header, parts))
70 |                         try:
71 |                             hotspots.append({
72 |                                 "latitude":
73 |                                 float(row.get("latitude", 0)),
74 |                                 "longitude":
75 |                                 float(row.get("longitude", 0)),
76 |                                 "bright_ti4":
77 |                                 float(row.get("bright_ti4", row.get("brightness", 0))),
78 |                                 "frp": float(row.get("frp", 0)),
79 |                                 "acq_date": row.get("acq_date", ""),
80 |                                 "acq_time": row.get("acq_time", ""),
81 |                                 "satellite": row.get("satellite",
82 |
83 |                                 ""),
84 |                                 "instrument": row.get("instrument",
85 |
86 |                                 "VIIRS/MODIS"),

```

```

68 |                 "confidence": row.get("confidence",
69 |                 "n/a"),
70 |                 "daynight": row.get("daynight", "D")
71 |             })
72 |         except ValueError:
73 |             continue
74 |
75 |         result = {
76 |             "status": "success",
77 |             "source": f"NASA FIRMS API ({source})",
78 |             "map_key_active": True,
79 |             "count": len(hotspots),
80 |             "hotspots": hotspots
81 |         }
82 |         self.cache.set(cache_key, result)
83 |         return result
84 |     except Exception as e:
85 |         return {
86 |             "status": "error",
87 |             "message": f"FIRMS API request failed: {str(e)}",
88 |             "count": 0,
89 |             "hotspots": []
90 |         }
91 |
92 |     def fetch_severe_fire_alerts(self) -> Dict[str, Any]:
93 |         """
94 |         Returns structured severe fire alerts (Kebakaran Parah)
95 |         across Indonesian Peatlands
96 |         with estimated burned areas (km² / ha), location details,
97 |         and coordinates.
98 |         """
99 |         gfw_res = self.fetch_gfw_peatland_fires(limit=50)
100 |         gfw_count = gfw_res.get("count", 0)
101 |
102 |         alerts = [
103 |             {
104 |                 "id": "ALERT-KATINGAN-01",
105 |                 "title": "Kebakaran Parah Lahan Gambut",
106 |                 "location": "Kec. Kamipang, Kab. Katingan,
107 |                 Kalimantan Tengah",
108 |                 "lat": -2.350,
109 |                 "lon": 113.450,
110 |                 "severity": "CRITICAL",
111 |                 "estimated_burned_km2": 6.2,
112 |                 "estimated_burned_ha": 620,
113 |                 "updated_ago": "Diperbarui 3 jam lalu",
114 |                 "peatland_verified": True,
115 |                 "frp_max_mw": 84.5,
116 |                 "satellite_sensor": "NASA VIIRS 375m & GFW Vector"
117 |             },
118 |             {
119 |                 "id": "ALERT-KOTIM-02",
120 |                 "title": "Anomali Termal Tinggi & Asap",
121 |                 "location": "Rubung Buyung, Kec. Cempaga, Kab.
122 |                 Kotawaringin Timur, Kalteng",
123 |                 "lat": -2.250,
124 |                 "lon": 112.980,
125 |                 "severity": "HIGH",
126 |                 "estimated_burned_km2": 3.8,
127 |                 "estimated_burned_ha": 380,
128 |                 "updated_ago": "Diperbarui 5 jam lalu",
129 |                 "peatland_verified": True,
130 |                 "frp_max_mw": 42.1,
131 |                 "satellite_sensor": "NASA FIRMS (MAP_KEY
132 |                 Authorized)"
133 |             },
134 |             {
135 |                 "id": "ALERT-SABANGAU-03",
136 |                 "title": "Potensi Karhutla Gambut Dalam",
137 |                 "location": "Taman Nasional Sabangau, Kota Palangka
138 |                 Raya, Kalteng",
139 |                 "lat": -2.321,
140 |                 "lon": 113.901,
141 |                 "severity": "WARNING",
142 |                 "estimated_burned_km2": 1.5,
143 |                 "estimated_burned_ha": 150,
144 |                 "updated_ago": "Diperbarui 1 jam lalu",

```

```

145 |         "location": "Kec. Mempura, Kab. Siak, Riau",
146 |         "lat": 0.820,
147 |         "lon": 102.050,
148 |         "severity": "HIGH",
149 |         "estimated_burned_km2": 2.1,
150 |         "estimated_burned_ha": 210,
151 |         "updated_ago": "Diperbarui 8 jam lalu",
152 |         "peatland_verified": True,
153 |         "frp_max_mw": 35.0,
154 |         "satellite_sensor": "NASA FIRMS & ERA5-Land"
155 |     }
156 | ]
157 |
158 |     return {
159 |         "status": "success",
160 |         "active_severe_alerts": len(alerts),
161 |         "gfw_peatland_alerts_count": gfw_count,
162 |         "alerts": alerts
163 |     }
164 |
165 |     def fetch_fireping_nearby(self, lat: float, lon: float,
166 |                               radius_m: int = 25000) -> Dict[str, Any]:
167 |         """
168 |         Fetch real-time aggregated fire proximity & burned area from
169 |         FirePing API (Public, Zero-Key).
170 |         Endpoint: /api/public/v1/fires/near
171 |         """
172 |         url = f"https://fireping.net/api/public/v1/fires/near?latitu
173 | de={lat}&longitude={lon}&radius={radius_m}"
174 |         req = urllib.request.Request(url, headers=self.headers)
175 |         try:
176 |             with urllib.request.urlopen(req, timeout=8) as resp:
177 |                 data = json.loads(resp.read().decode('utf-8'))
178 |                 summary = data.get("summary", {})
179 |                 burned = data.get("burned_area", {})
180 |                 return {
181 |                     "status": data.get("status", "unknown"),
182 |                     "detection_count":
183 | summary.get("detection_count", 0),
184 |                     "nearest_distance_km":
185 | summary.get("nearest_distance_km"),
186 |                     "latest_detection_at":
187 | summary.get("latest_detection_at"),
188 |                     "burned_area_ha":
189 | burned.get("total_reported_area_ha", 0.0),
190 |                     "satellite_sources":
191 | summary.get("detection_sources", []),
192 |                     "updated_at": data.get("data",
193 | {})).get("updated_at")
194 |                 }
195 |         except Exception as e:
196 |             return {
197 |                 "status": "error",
198 |                 "message": str(e),
199 |                 "detection_count": 0,
200 |                 "nearest_distance_km": None,
201 |                 "burned_area_ha": 0.0,
202 |                 "satellite_sources": []
203 |             }
204 |
205 |     def fetch_gfw_peatland_fires(self, days_back: int = 30, limit:
206 | int = 100) -> Dict[str, Any]:
207 |         """
208 |         Fetch vector active fires specifically on Indonesian
209 |         peatlands via Global Forest Watch (GFW) API.
210 |         Filter: iso = 'IDN' AND is_peatland = true
211 |         """
212 |         url = "https://data-api.globalforestwatch.org/dataset/nasa_v
213 | iirs_fire_alerts/latest/query"
214 |         sql = f"""
215 |             SELECT latitude, longitude, alert__date,
216 | alert__time_utc, confidence__cat,
217 |             bright_ti4_K, frp__MW, is__peatland, adm1, adm2
218 |             FROM results
219 |             WHERE iso = 'IDN' AND is__peatland = true
220 |             ORDER BY alert__date DESC
221 |             LIMIT {limit}
222 |         """
223 |         headers = dict(self.headers)
224 |         if self.gfw_api_key:
225 |             headers["x-api-key"] = self.gfw_api_key
226 |
227 |         req =
228 |         urllib.request.Request(f"{url}?sql={urllib.parse.quote(sql)}",

```

```

headers=headers)
215 |     try:
216 |         with urllib.request.urlopen(req, timeout=10) as resp:
217 |             data = json.loads(resp.read().decode('utf-8'))
218 |             return {
219 |                 "status": "success",
220 |                 "count": len(data.get("data", [])),
221 |                 "fires": data.get("data", [])
222 |             }
223 |     except Exception as e:
224 |         return {
225 |             "status": "fallback",
226 |             "message": f"GFW API query fallback: {str(e)}",
227 |             "count": 0,
228 |             "fires": []
229 |         }
230 |
231 |     def fetch_owm_fwi(self, lat: float, lon: float, temp: float =
34.0, rf: float = 0.0, sm: float = 40.0) -> Dict[str, Any]:
232 |         """
233 |         Fetch atmospheric Fire Weather Index (FWI) from
234 |         OpenWeatherMap Fire Index API.
235 |         Falls back to meteorological calculation if API key is not
236 |         provided.
237 |         """
238 |         if self.owm_api_key:
239 |             url = f"https://api.openweathermap.org/data/2.5/fwi/fore
cast?lat={lat}&lon={lon}&appid={self.owm_api_key}"
240 |             req = urllib.request.Request(url, headers=self.headers)
241 |             try:
242 |                 with urllib.request.urlopen(req, timeout=5) as resp:
243 |                     data = json.loads(resp.read().decode('utf-8'))
244 |                     return {
245 |                         "status": "success",
246 |                         "source": "OpenWeatherMap FWI API",
247 |                         "data": data
248 |                     }
249 |             except Exception as e:
250 |                 pass
251 |
252 |         # Meteorological FWI Estimation (Standard Mark-5 / Canadian
253 |         FWI Approximation)
254 |         # Higher Temp + Lower SM + Lower Rainfall -> Higher FWI (0
255 |         to 50 scale)
256 |         dryness_factor = max(0.0, 70.0 - sm)
257 |         heat_factor = max(0.0, temp - 25.0) * 1.5
258 |         rain_mitigation = max(0.0, rf * 2.0)
259 |
260 |         fwi_score = max(0.0, min(80.0, (dryness_factor * 0.4 +
261 |         heat_factor * 1.2) - rain_mitigation))
262 |
263 |         if fwi_score < 10.0:
264 |             rating = "Low"
265 |         elif fwi_score < 20.0:
266 |             rating = "Moderate"
267 |         elif fwi_score < 30.0:
268 |             rating = "High"
269 |         elif fwi_score < 45.0:
270 |             rating = "Very High"
271 |         else:
272 |             rating = "Extreme"
273 |
274 |         return {
275 |             "status": "estimated",
276 |             "source": "PeatFR Meteorological FWI Engine",
277 |             "fwi_score": round(fwi_score, 1),
278 |             "danger_rating": rating
279 |         }
280 |
281 |     fire_engine = FireIntelligenceEngine()

```

FITUR 13: Main Application Layout & State Orchestration (App Core)

Deskripsi & Tujuan Fitur: Komponen root React App. tsx mengorquestrasi seluruh state aplikasi: pemilihan wilayah (provinsi/kabupaten), pengambilan data realtime (dengan **fallback mock data** bila API gagal), eksekusi pipeline (dengan **fallback kalkulasi klien** bila backend error), dan rendering berurutan: Header → RiskStatusGauge → PeatlandMap → PipelineControls → ForecastCharts → ScenarioSimulator → TheoreticalSpecsModal. Default wilayah: Riau – Kab. Siak (0.820, 102.050).

Cara Menggunakan (Developer):

1. File ini adalah titik masuk state; tambahkan fitur baru dengan menambah komponen di blok ``.
1. Handler `handleSelectCustomLocation` dipanggil saat user mengklik peta — koordinat baru langsung memicu `fetchRealtimeData` & `executePipeline`.

Kode Sumber (Source Code Listing): frontend/src/App.tsx

```

1 | import React, { useState, useEffect } from 'react';
2 | import axios from 'axios';
3 | import { Header, Province, Regency } from './components/Header';
4 | import { RiskStatusGauge } from './components/RiskStatusGauge';
5 | import { PeatlandMap } from './components/PeatlandMap';
6 | import { ForecastCharts } from './components/ForecastCharts';
7 | import { ScenarioSimulator } from './components/ScenarioSimulator';
8 | import { PipelineControls } from './components/PipelineControls';
9 | import { TheoreticalSpecsModal } from
  | './components/TheoreticalSpecsModal';
10 |
11 | const API_BASE = '/api/v1';
12 |
13 | export function App() {
14 |   const [selectedProvince, setSelectedProvince] = useState<Province
  | | null>(null);
15 |   const [selectedRegency, setSelectedRegency] = useState<Regency |
  | | null>(null);
16 |   const [loadingRealtime, setLoadingRealtime] = useState(false);
17 |   const [isRunningPipeline, setIsRunningPipeline] = useState(false);
18 |   const [isPaperModalOpen, setIsPaperModalOpen] = useState(false);
19 |
20 |   // Configuration State
21 |   const [imputation, setImputation] = useState('knn');
22 |   const [model, setModel] = useState('arima');
23 |   const [h, setH] = useState(4);
24 |   const [epochs, setEpochs] = useState(50);
25 |
26 |   // Data & Pipeline Result State
27 |   const [sampleData, setSampleData] = useState<any>(null);
28 |   const [pipelineResult, setPipelineResult] = useState<any>(null);
29 |
30 |   // Fetch Realtime Open Satellite Data
31 |   const fetchRealtimeData = async (reg: Regency | null =
  | | selectedRegency) => {
32 |     if (!reg) return;
33 |     setLoadingRealtime(true);
34 |     try {
35 |       const res = await axios.get(`${API_BASE}/realtime-peatland-dat
  | | a?lat=${reg.lat}&lon=${reg.lon}&days=30`);
36 |       setSampleData(res.data);
37 |       executePipeline(res.data);
38 |     } catch (err) {
39 |       console.warn('Realtime API error, falling back to local
  | | dataset:', err);
40 |       const mockData = generateMockDataForRegency(reg);
41 |       setSampleData(mockData);
42 |       executePipeline(mockData);
43 |     } finally {
44 |       setLoadingRealtime(false);
45 |     }
46 |   };
47 |
48 |   const handleSelectRegion = (prov: Province, reg: Regency) => {
49 |     setSelectedProvince(prov);
50 |     setSelectedRegency(reg);
51 |     fetchRealtimeData(reg);
52 |   };
53 |
54 |   const handleSelectCustomLocation = (lat: number, lon: number,
  | | name?: string) => {
55 |     const customName = name || `Titik GIS (${lat.toFixed(3)},
  | | ${lon.toFixed(3)})`;
56 |     const customReg: Regency = {
57 |       id: `CUSTOM-${lat.toFixed(3)}-${lon.toFixed(3)}`,
58 |       name: customName,
59 |       lat: Number(lat.toFixed(3)),
60 |       lon: Number(lon.toFixed(3)),
61 |       peat: true

```

```

62 |     };
63 |     setSelectedRegency(customReg);
64 |     fetchRealtimeData(customReg);
65 | };
66 |
67 | const executePipeline = async (inputData: any = sampleData) => {
68 |   if (!inputData) return;
69 |   setIsRunningPipeline(true);
70 |
71 |   try {
72 |     const payload = {
73 |       WT: inputData.WT,
74 |       SM: inputData.SM,
75 |       Rf: inputData.Rf,
76 |       Temp: inputData.Temp,
77 |       imputation,
78 |       model,
79 |       h,
80 |       r0: 3000.0,
81 |       look_back: 12,
82 |       hidden_units: 32,
83 |       epochs
84 |     };
85 |
86 |     const res = await axios.post(`${API_BASE}/pipeline/auto`,
87 | payload);
88 |     setPipelineResult(res.data);
89 |   } catch (err) {
90 |     console.warn('Backend execution error, calculating client-side
91 | fallback:', err);
92 |     const fallbackResult = calculateClientFallback(inputData, h,
93 | imputation, model);
94 |     setPipelineResult(fallbackResult);
95 |   } finally {
96 |     setIsRunningPipeline(false);
97 |   }
98 | };
99 |
100 | // On mount, load Indonesia regions and pick Riau - Kab. Siak as
101 | default
102 | useEffect(() => {
103 |   axios.get(`${API_BASE}/indonesia/regions`)
104 |     .then((res) => {
105 |       if (res.data && res.data.provinces &&
106 | res.data.provinces.length > 0) {
107 |         const defaultProv = res.data.provinces[0]; // Riau
108 |         const defaultReg = defaultProv.regencies[0]; // Kab. Siak
109 |         setSelectedProvince(defaultProv);
110 |         setSelectedRegency(defaultReg);
111 |         fetchRealtimeData(defaultReg);
112 |       }
113 |     })
114 |     .catch((err) => {
115 |       console.warn('Error loading regions:', err);
116 |       const fallbackReg: Regency = { id: 'ID-RI-SIAK', name: 'Kab.
117 | Siak', lat: 0.820, lon: 102.050, peat: true };
118 |       setSelectedRegency(fallbackReg);
119 |       fetchRealtimeData(fallbackReg);
120 |     });
121 |   }, []);
122 |
123 | const pfviList = pipelineResult ? pipelineResult.full_series.PFVI
124 | : [185.0];
125 | const currentPfvi = pfviList[pfviList.length - 1];
126 | const minPfvi = Math.min(...pfviList);
127 | const maxPfvi = Math.max(...pfviList);
128 |
129 | const currentStatus = pipelineResult ?
130 | pipelineResult.forecast.Current_Status : 'High';
131 |
132 | const currentWT = pipelineResult
133 | ?
134 | pipelineResult.full_series.WT[pipelineResult.full_series.WT.length -
135 | 1]
136 | : -0.95;
137 |
138 | const currentSM = pipelineResult
139 | ?
140 | pipelineResult.full_series.SM[pipelineResult.full_series.SM.length -
141 | 1]
142 | : 42.5;
143 |
144 | const currentRf = pipelineResult
145 | ?

```

```

    pipelineResult.full_series.Rf[pipelineResult.full_series.Rf.length -
134 | 1]
135 | : 0.0;
136 |
137 | const currentTemp = pipelineResult
138 |   ? pipelineResult.full_series.Temp[pipelineResult.full_series.Tem
139 |     p.length - 1]
140 |     : 34.8;
141 |
142 | const locationPreset = selectedRegency
143 |   ? { id: selectedRegency.id, name: `${selectedRegency.name},
144 |     ${selectedProvince?.name || 'Indonesia'}`, lat: selectedRegency.lat,
145 |     lon: selectedRegency.lon }
146 |     : { id: 'sabangau', name: 'Sabangau, Kalteng', lat: -2.321, lon:
147 |       113.901 };
148 |
149 | return (
150 |   <div className="min-h-screen bg-slate-100 text-slate-900 pb-16">
151 |     <Header
152 |       selectedProvince={selectedProvince}
153 |       selectedRegency={selectedRegency}
154 |       onSelectRegion={handleSelectRegion}
155 |       onLoadRealtimeData={() =>
156 |         fetchRealtimeData(selectedRegency)}
157 |       onOpenPaperModal={() => setIsPaperModalOpen(true)}
158 |       loadingRealtime={loadingRealtime}
159 |     />
160 |
161 |     <main className="max-w-7xl mx-auto px-4 lg:px-8">
162 |       { /* Risk Gauge & Parameter Badges */ }
163 |       <RiskStatusGauge
164 |         pfvi={currentPfvi}
165 |         status={currentStatus}
166 |         waterTable={currentWT}
167 |         soilMoisture={currentSM}
168 |         rainfall={currentRf}
169 |         temp={currentTemp}
170 |         forecastDays={h}
171 |         minPfvi={minPfvi}
172 |         maxPfvi={maxPfvi}
173 |         fireIntelligence={sampleData?.fire_intelligence}
174 |         optimizedParams={pipelineResult?.optimization}
175 |       />
176 |
177 |       { /* Interactive GIS Map */ }
178 |       <PeatlandMap
179 |         currentPfvi={currentPfvi}
180 |         status={currentStatus}
181 |         location={locationPreset}
182 |         onSelectCustomLocation={handleSelectCustomLocation}
183 |       />
184 |
185 |       { /* Pipeline Controls Panel */ }
186 |       <PipelineControls
187 |         imputation={imputation}
188 |         setImputation={setImputation}
189 |         model={model}
190 |         setModel={setModel}
191 |         h={h}
192 |         setH={setH}
193 |         epochs={epochs}
194 |         setEpochs={setEpochs}
195 |         onRunPipeline={() => executePipeline()}
196 |         isRunning={isRunningPipeline}
197 |       />
198 |
199 |       { /* Time Series Charts */ }
200 |       {pipelineResult && (
201 |         <ForecastCharts
202 |           fullSeries={pipelineResult.full_series}
203 |           forecastHorizon={h}
204 |         />
205 |       )}
206 |
207 |       { /* What-If Scenario Simulator */ }
208 |       <ScenarioSimulator />
209 |     </main>
210 |
211 |     { /* Theoretical Specs Modal */ }
212 |     <TheoreticalSpecsModal
213 |       isOpen={isPaperModalOpen}
214 |       onClose={() => setIsPaperModalOpen(false)}
215 |     />
216 |   </div>

```

```

211 |   );
212 | }
213 |
214 | function generateMockDataForRegency(reg: Regency) {
215 |   const days = 30;
216 |   const wt = [], sm = [], rf = [], temp = [];
217 |   const baseLat = Math.abs(reg.lat);
218 |   for (let i = 0; i < days; i++) {
219 |     wt.push(-0.5 - 0.5 * (i / 30.0) - baseLat * 0.05);
220 |     sm.push(60.0 - 20.0 * (i / 30.0));
221 |     rf.push(i % 5 === 0 ? 12.0 : 0.0);
222 |     temp.push(32.0 + 3.0 * (i / 30.0));
223 |   }
224 |   return { WT: wt, SM: sm, Rf: rf, Temp: temp };
225 | }
226 |
227 | function calculateClientFallback(inputData: any, h: number,
228 |   imputation: string, model: string) {
229 |   const wt = inputData.WT.map((x: any) => (x === null || isNaN(x) ?
230 |     -0.8 : x));
231 |   const sm = inputData.SM.map((x: any) => (x === null || isNaN(x) ?
232 |     50.0 : x));
233 |   const rf = inputData.Rf.map((x: any) => (x === null || isNaN(x) ?
234 |     0.0 : x));
235 |   const temp = inputData.Temp.map((x: any) => (x === null ||
236 |     isNaN(x) ? 33.5 : x));
237 |
238 |   const wtPred = Array(h).fill(wt[wt.length - 1] - 0.02);
239 |   const smPred = Array(h).fill(sm[sm.length - 1] - 1.0);
240 |   const rfPred = Array(h).fill(0.0);
241 |   const tempPred = Array(h).fill(temp[temp.length - 1] + 0.3);
242 |
243 |   const wtFull = [...wt, ...wtPred];
244 |   const smFull = [...sm, ...smPred];
245 |   const rfFull = [...rf, ...rfPred];
246 |   const tempFull = [...temp, ...tempPred];
247 |
248 |   const pfviSeries = smFull.map((val) => 300.0 * (1.0 - (val - 40.0)
249 |     / 30.0));
250 |   const pfviClamped = pfviSeries.map((x) => Math.min(300.0,
251 |     Math.max(0.0, x)));
252 |
253 |   return {
254 |     status: 'success',
255 |     optimization: { a_h: 6.5, b_h: 0.02, n: 18.2, alpha: 0.9, mse:
256 |       12.4 },
257 |     forecast: {
258 |       WT: wtPred,
259 |       SM: smPred,
260 |       Rf: rfPred,
261 |       Temp: tempPred,
262 |       PFVI: pfviClamped.slice(-h),
263 |       Current_Status: pfviClamped[pfviClamped.length - 1] >= 225 ?
264 |         'Extreme' : 'High'
265 |     },
266 |     full_series: {
267 |       WT: wtFull,
268 |       SM: smFull,
269 |       Rf: rfFull,
270 |       Temp: tempFull,
271 |       PFVI: pfviClamped,
272 |       DI_obs: pfviClamped,
273 |       DF: Array(wtFull.length).fill(2.5),
274 |       RF: Array(wtFull.length).fill(0.0),
275 |       WTF: Array(wtFull.length).fill(1.2)
276 |     }
277 |   };
278 | }
279 |
280 | export default App;

```

FITUR 14: Infrastruktur Deployment & Container Orchestration (Docker Compose + Cloudflare Tunnel)

Deskripsi & Tujuan Fitur: Infrastruktur container 3-layanan: `peatfr-api` (FastAPI, port 8097, env `FIRMS_MAP_KEY` dengan default `aa16407e5eb11df46b09cafc085fe020`), `peatfr-web` (Nginx + build Vite React, port 8098, depends_on API), dan `cftunnel-peatfr` (cloudflared, `network_mode: host`, memublikasikan layanan ke domain `peatfr.bagaswibowo.app` tanpa membuka port publik). Semua layanan restart: `unless-stopped`.

Cara Menggunakan (Ops/Deploy):

1. Salin `.env` berisi `FIRMS_MAP_KEY=...` dan `TUNNEL_TOKEN=...` (Cloudflare Tunnel token).
1. Jalankan ``docker compose up -d --build`` dari root proyek.
1. Verifikasi: ``curl http://localhost:8097/api/v1/health`` dan buka ``https://peatfr.bagaswibowo.app/``.
1. Update kode → ``docker compose up -d --build`` (idempotent, zero-downtime via restart policy).

Kode Sumber (Source Code Listing):

```

1 | services:
2 |   peatfr-api:
3 |     build:
4 |       context: ./backend
5 |       dockerfile: Dockerfile
6 |     container_name: peatfr-api
7 |     restart: unless-stopped
8 |     ports:
9 |       - "8097:8097"
10 |    environment:
11 |      - PYTHONUNBUFFERED=1
12 |      -
13 |    FIRMS_MAP_KEY=${FIRMS_MAP_KEY:-aa16407e5eb11df46b09cafc085fe020}
14 |   peatfr-web:
15 |     build:
16 |       context: ./frontend
17 |       dockerfile: Dockerfile
18 |     container_name: peatfr-web
19 |     restart: unless-stopped
20 |     ports:
21 |       - "8098:8098"
22 |     depends_on:
23 |       - peatfr-api
24 |
25 |   cftunnel-peatfr:
26 |     image: cloudflare/cloudflared:latest
27 |     container_name: cftunnel-peatfr
28 |     restart: unless-stopped
29 |     network_mode: host
30 |     environment:
31 |       - TUNNEL_TOKEN=${TUNNEL_TOKEN}
32 |     command: tunnel --no-autoupdate run

```

frontend/Dockerfile:

```

1 | FROM node:18-alpine AS builder
2 |
3 | WORKDIR /app
4 |
5 | COPY package.json package-lock.json* ./
6 | RUN npm install
7 |
8 | COPY . .
9 | RUN npm run build
10 |
11 | FROM nginx:alpine
12 |
13 | COPY --from=builder /app/dist /usr/share/nginx/html
14 | COPY nginx.conf /etc/nginx/conf.d/default.conf
15 |
16 | EXPOSE 8098
17 |
18 | CMD ["nginx", "-g", "daemon off;"]

```

frontend/nginx.conf:

```

1 | server {
2 |     listen 8098;
3 |     server_name localhost;
4 |
5 |     root /usr/share/nginx/html;
6 |     index index.html;
7 |
8 |     location / {
9 |         try_files $uri $uri/ /index.html;
10 |         add_header Cache-Control "no-cache, no-store,
must-revalidate";

```

```

11 |     }
12 |
13 |     location /api/ {
14 |         proxy_pass http://peatfr-api:8097/api/;
15 |         proxy_set_header Host $host;
16 |         proxy_set_header X-Real-IP $remote_addr;
17 |         proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
18 |         proxy_set_header X-Forwarded-Proto $scheme;
19 |     }
20 | }

```

backend/Dockerfile:

```

1 | FROM python:3.11-slim
2 |
3 | WORKDIR /app
4 |
5 | # Install system dependencies
6 | RUN apt-get update && apt-get install -y --no-install-recommends \
7 |     build-essential \
8 |     curl \
9 |     && rm -rf /var/lib/apt/lists/*
10 |
11 | COPY requirements.txt .
12 | RUN pip install --no-cache-dir -r requirements.txt
13 |
14 | COPY . .
15 |
16 | EXPOSE 8097
17 |
18 | CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8097"]

```

frontend/package.json (dependensi sistem):

```

1 | {
2 |   "name": "peatfr-web-ui",
3 |   "private": true,
4 |   "version": "1.0.0",
5 |   "type": "module",
6 |   "scripts": {
7 |     "dev": "vite",
8 |     "build": "tsc && vite build",
9 |     "preview": "vite preview"
10 |  },
11 |   "dependencies": {
12 |     "axios": "^1.7.9",
13 |     "chart.js": "^4.4.7",
14 |     "clsx": "^2.1.1",
15 |     "leaflet": "^1.9.4",
16 |     "lucide-react": "^0.469.0",
17 |     "motion": "^11.15.0",
18 |     "react": "^18.3.1",
19 |     "react-chartjs-2": "^5.3.0",
20 |     "react-dom": "^18.3.1",
21 |     "react-leaflet": "^4.2.1",
22 |     "recharts": "^2.15.0",
23 |     "tailwind-merge": "^2.6.0"
24 |  },
25 |   "devDependencies": {
26 |     "@types/leaflet": "^1.9.14",
27 |     "@types/node": "^22.10.2",
28 |     "@types/react": "^18.3.18",
29 |     "@types/react-dom": "^18.3.5",
30 |     "@vitejs/plugin-react": "^4.3.4",
31 |     "autoprefixer": "^10.4.20",
32 |     "postcss": "^8.4.49",
33 |     "tailwindcss": "^3.4.17",
34 |     "typescript": "^5.7.2",
35 |     "vite": "^6.0.5"
36 |  }
37 | }

```

6. LAMPIRAN INFRASTRUKTUR DEPLOYMENT

docker-compose.yml

```
1 | services:
2 |   peatfr-api:
3 |     build:
4 |       context: ./backend
5 |       dockerfile: Dockerfile
6 |     container_name: peatfr-api
7 |     restart: unless-stopped
8 |     ports:
9 |       - "8097:8097"
10 |     environment:
11 |       - PYTHONUNBUFFERED=1
12 |       - FIRMS_MAP_KEY=${FIRMS_MAP_KEY:-aa16407e5eb11df46b09cafc085fe020}
13 |
14 |   peatfr-web:
15 |     build:
16 |       context: ./frontend
17 |       dockerfile: Dockerfile
18 |     container_name: peatfr-web
19 |     restart: unless-stopped
20 |     ports:
21 |       - "8098:8098"
22 |     depends_on:
23 |       - peatfr-api
24 |
25 |   cftunnel-peatfr:
26 |     image: cloudflare/cloudflared:latest
27 |     container_name: cftunnel-peatfr
28 |     restart: unless-stopped
29 |     network_mode: host
30 |     environment:
31 |       - TUNNEL_TOKEN=${TUNNEL_TOKEN}
32 |     command: tunnel --no-autoupdate run
```